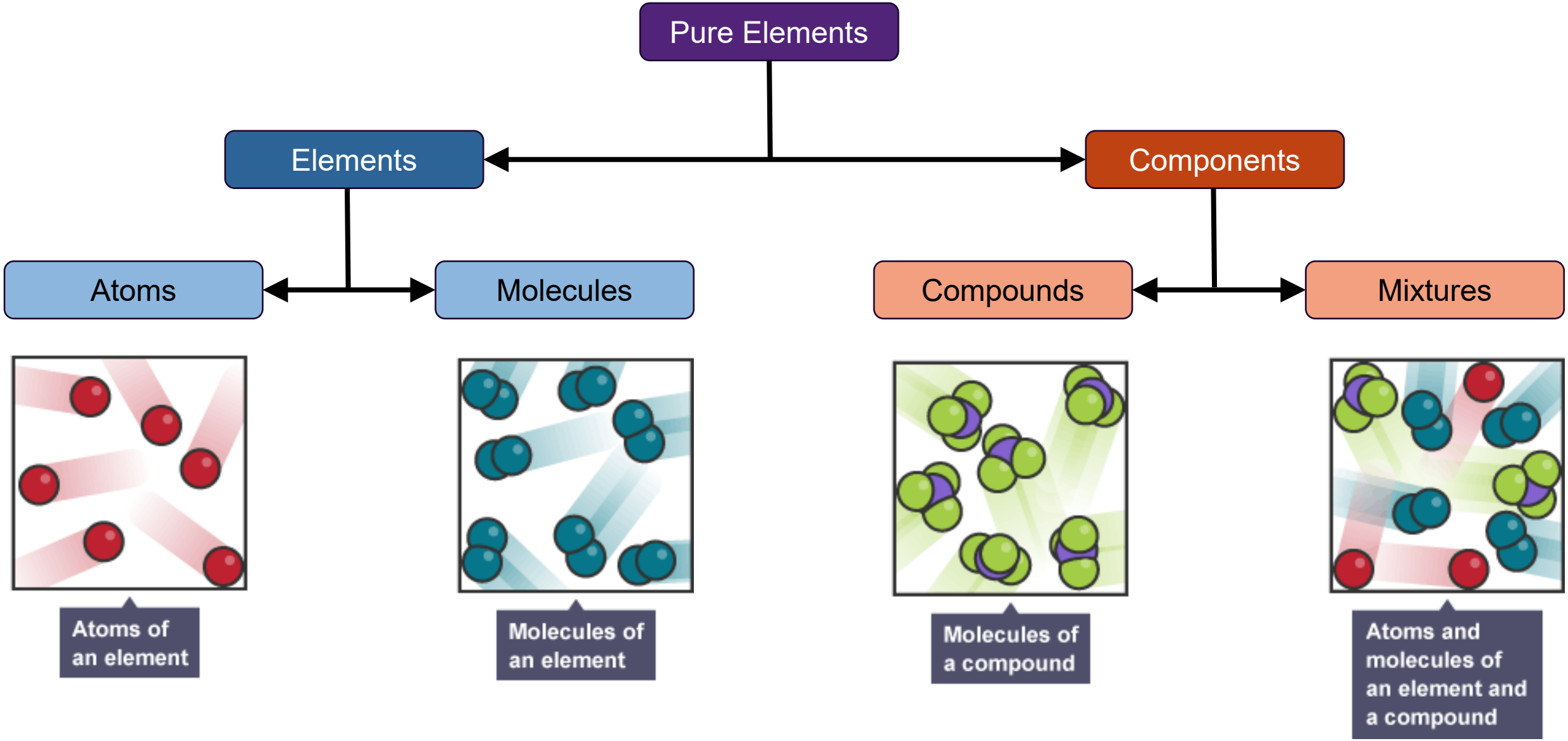
THE UNIVERSITY OF QUEENSLAND
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# MATE 6301

## X-ray Diffraction (XRD)

Lecture by,
Dr. Aditya Ashok (UQ-AIBN)

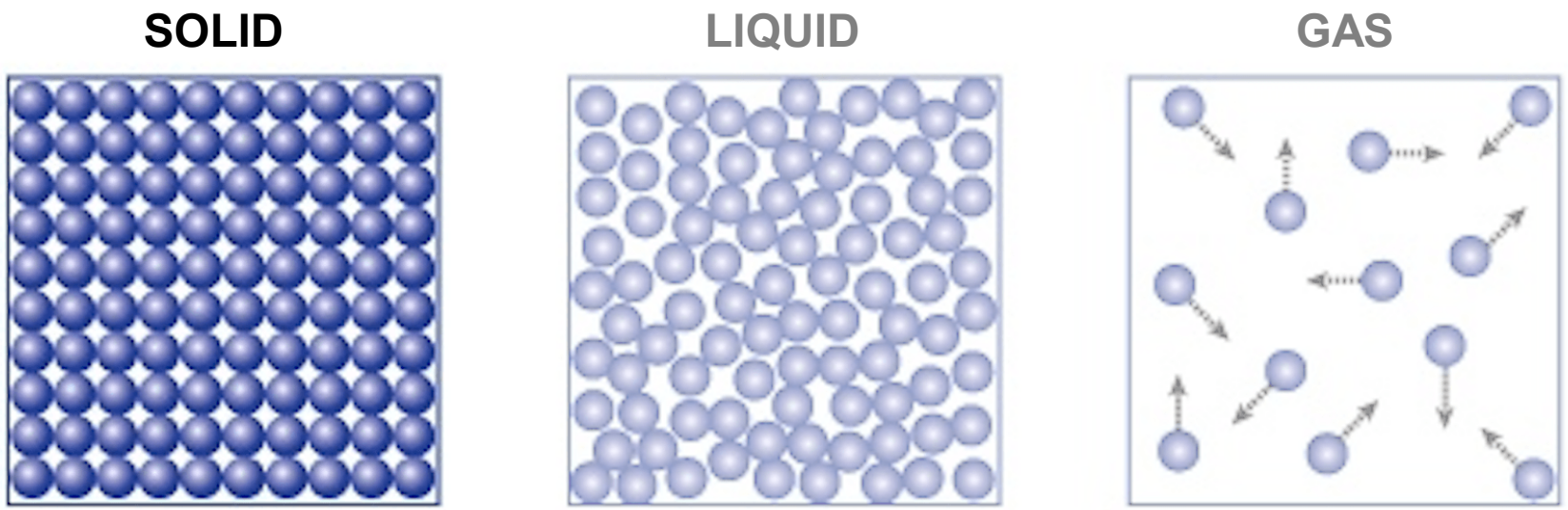

Pure Elements
Elements
Components
Atoms
Molecules
Compounds
Mixtures
Atoms of an element
Molecules of an element
Molecules of a compound
Atoms and molecules of an element and a compound

There are 3 primary states of matter,

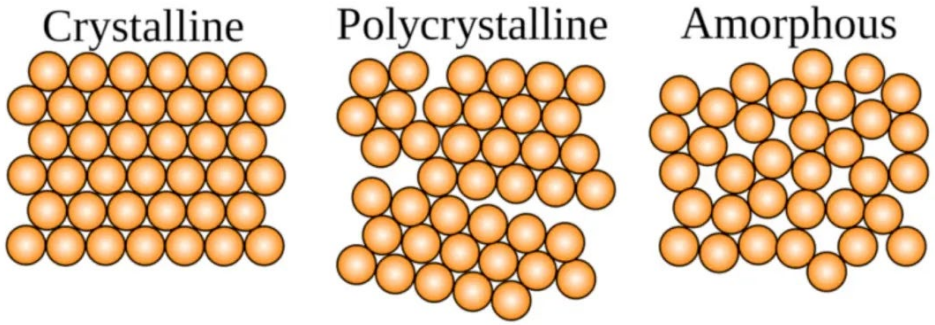
**Solid** – Matter with a fixed shape and volume. Particles are tightly packed and only vibrate in place (e.g., ice, rock).

**Liquid** – Matter with a fixed volume but no fixed shape. It takes the shape of its container. Particles are close but can move past each other (e.g., water, oil).

**Gas** – Matter with neither fixed shape nor fixed volume. Particles are far apart and move freely (e.g., air, steam).



Based on the arrangement of atoms,




i. Single Crystalline (long-range order)

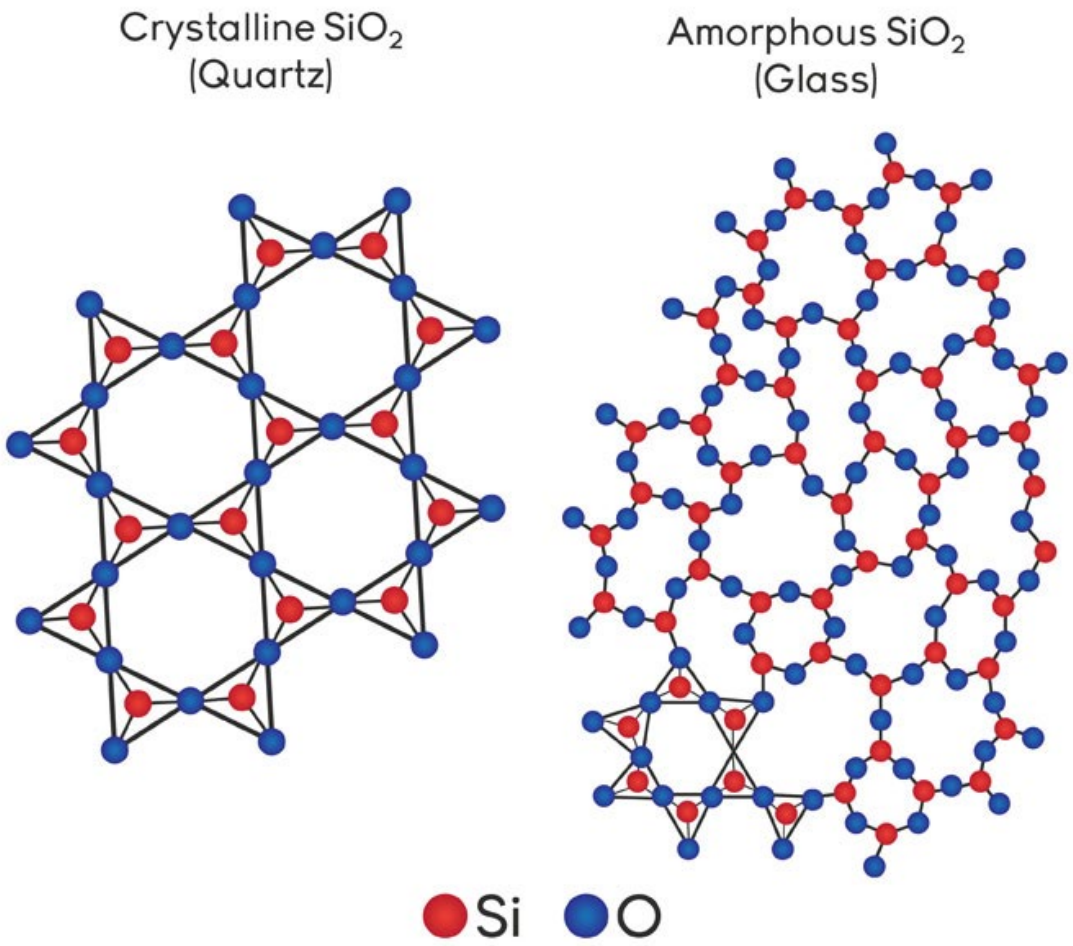
ii. Poly-crystalline (short-range order)

   a. Micro-crystalline

   b. Nano-crystalline

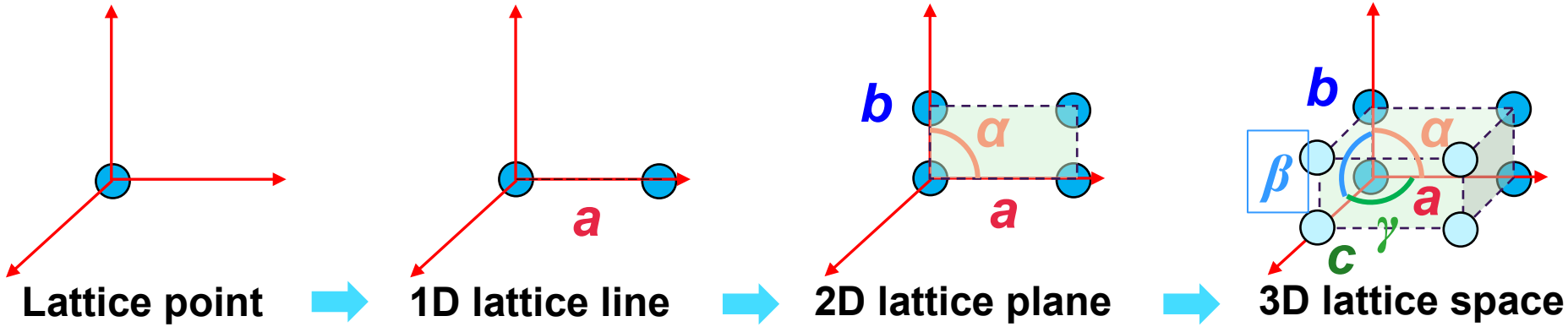
iii. Amorphous

Best example is silica glass,

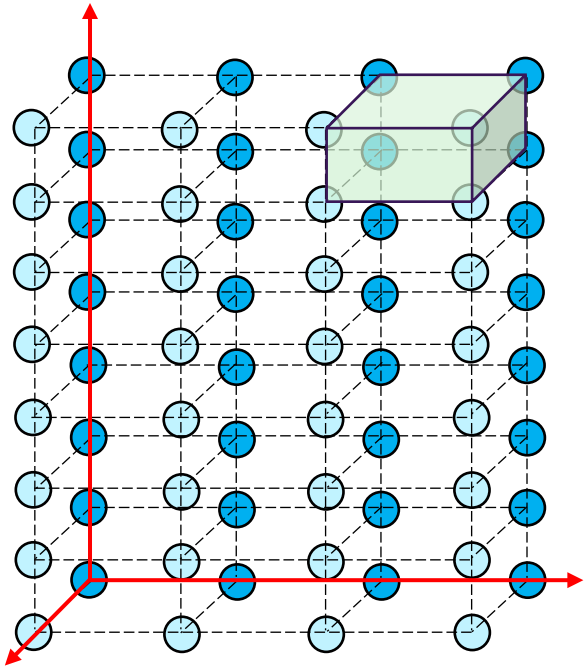


## What are crystalline state?

A crystal consists of identical *structural units*, consisting of one or more atoms, which are **regularly** and **periodically** arranged in three dimensions. This regularity extends over the entire crystal.




**Lattice point** → **1D lattice line** → **2D lattice plane** → **3D lattice space**

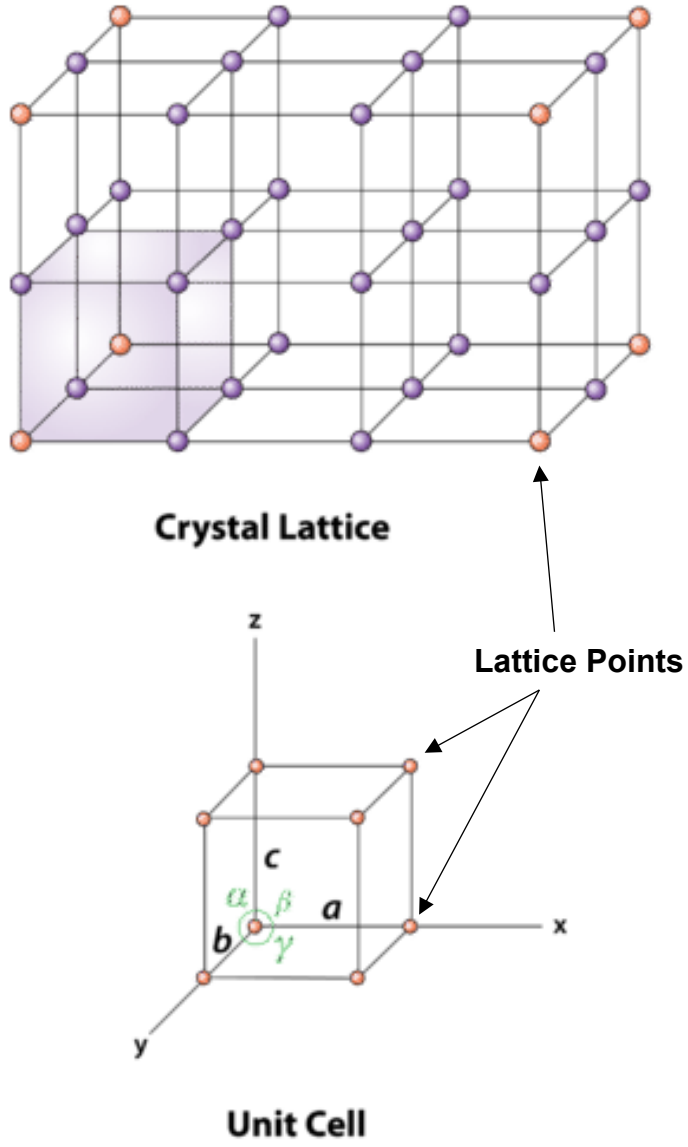


**Note:**
The smallest building blocks of a crystal are referred to as the **basis**, the periodic arrangement is described by the **lattice**.

## Crystal lattices and unit cells

Lattice is infinite periodic arrangement of points in a space.

The lattice is a 3D array of points (lattice points), each of them has identical surroundings.



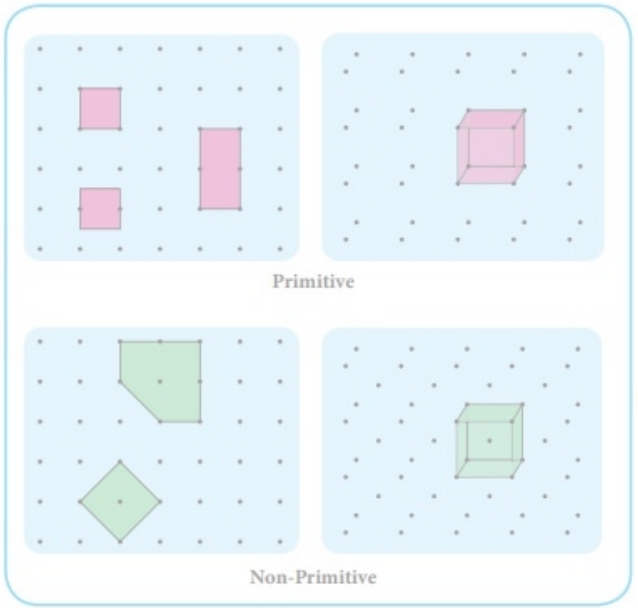

### Notes on Unit Cells:

- A **Unit Cell** represents a small volume of a lattice.
- The entire lattice can be reconstructed by repeating this unit cell **without empty space in-between.**
- The unit cell is defined by three independent unit vectors, **a**, **b**, and **c.**
- The direction of the unit vectors **a**, **b**, and **c** defines the direction of the *x*, *y* and *z* axes, called the **crystal axes.**
- α, β, γ are angles between **b** & **c**, **c** & **a**, and **a** & **b.**
- Maximum 6 variables.

## Primitive and non-primitive un

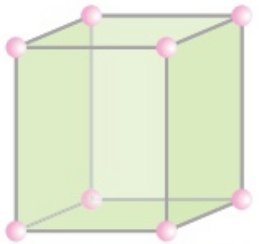
There are two types of unit c
primitive. A unit cell that conta
is called a primitive unit cell, wh
lattice points at each of the c
**present translational symme**

In case of non-primitive unit c
lattice points, either on a face
the unit cell.

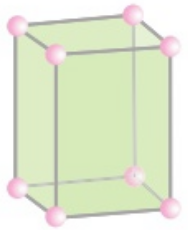



| | | | | |
|---|---|---|---|---|
| cubic | primitive | body centered | face centered | |
| tetragonal | primitive | body centered | | |
| hexagonal | | | | |
| orthorhombic | primitive | body centered | face centered | basis face centered |
| monoclinic | primitive | | basis face centered | |
| trigonal | | | | |
| triclinic | | | | |

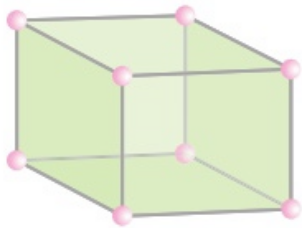
Table 6.2 14 Bravais Lattices



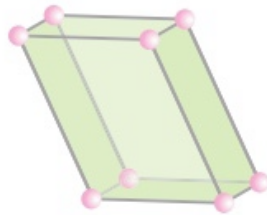
Hexagonal
$a = b \neq c$
$\alpha = \beta = 90°, \gamma = 120°$



Tetragonal
$a = b \neq c$
$\alpha = \beta = \gamma = 90°$



Monoclinic
$a \neq b \neq c$
$\alpha = \gamma = 90°, \beta \neq 90°$



Triclinic
$a \neq b \neq c$
$\alpha \neq \beta \neq \gamma \neq 90°$

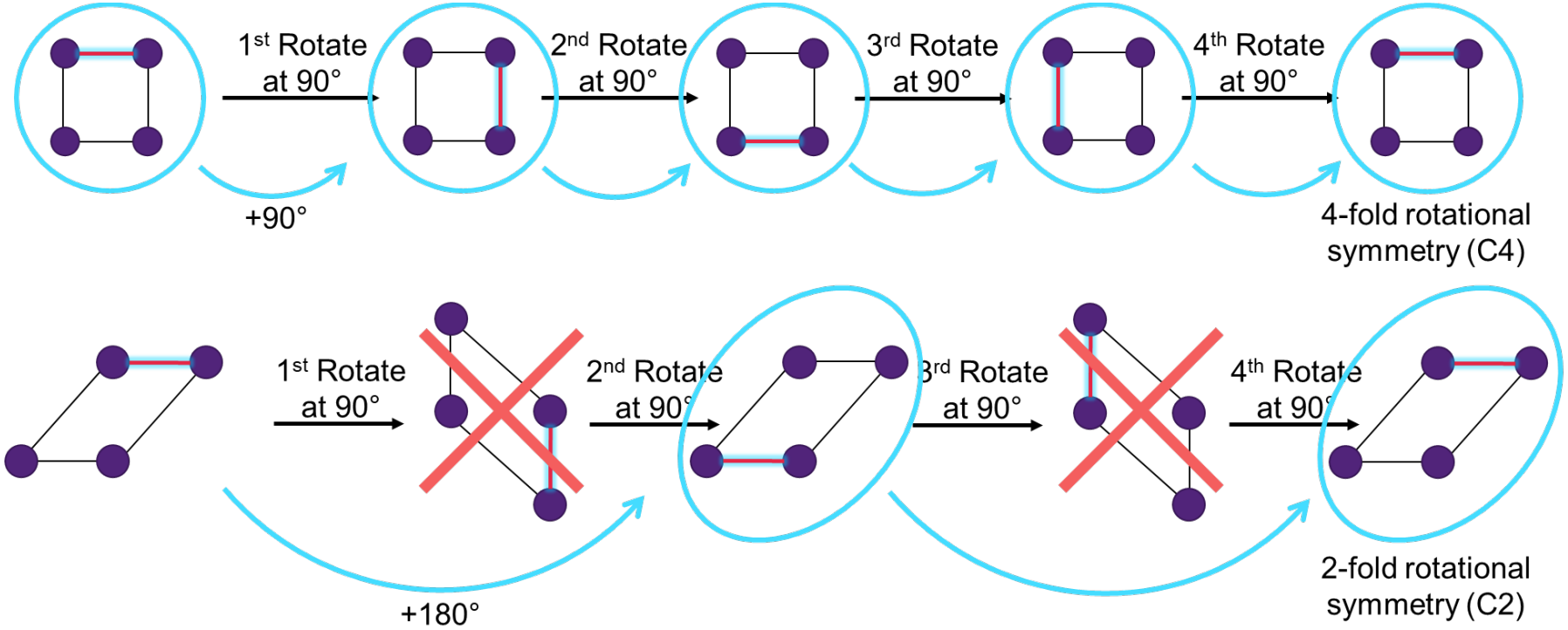
ɔic, tetragonal, orthorhombic,
They differ in the arrangement
ɔonding to the above seven,
nown in the figure.

Therefore, the basic rules of a primitive unit cells of 2D lattices,

**Rule #1:** **Unit cells should cover up the entire lattice by simple translation (*i.e.,* translational symmetry).**

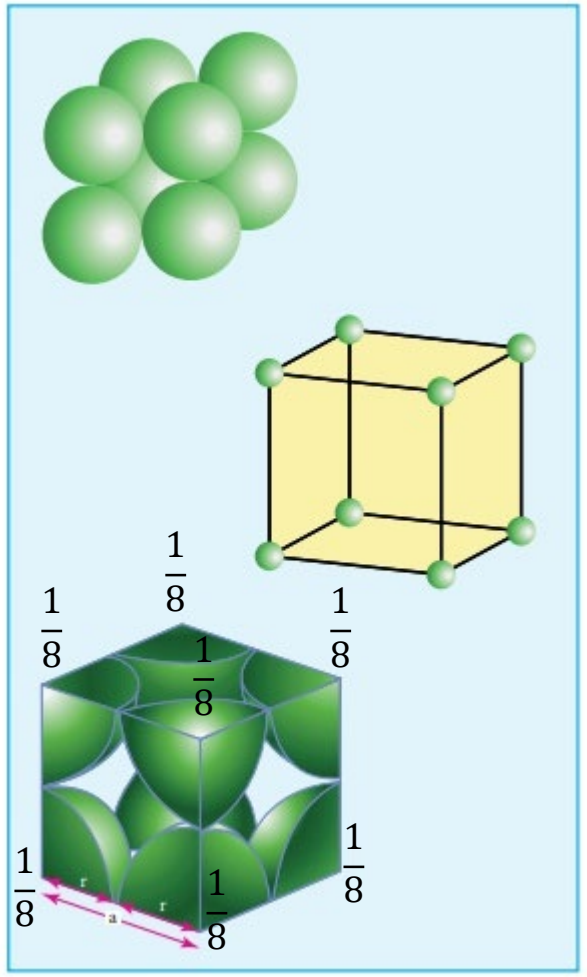
**Rule #2:** **Primitive unit cells should have lattice points only at the corners**

**Rule #3:** **Primitive unit cells best describe the symmetry of overall lattice (i.e., smallest rotational angle for symmetry).**



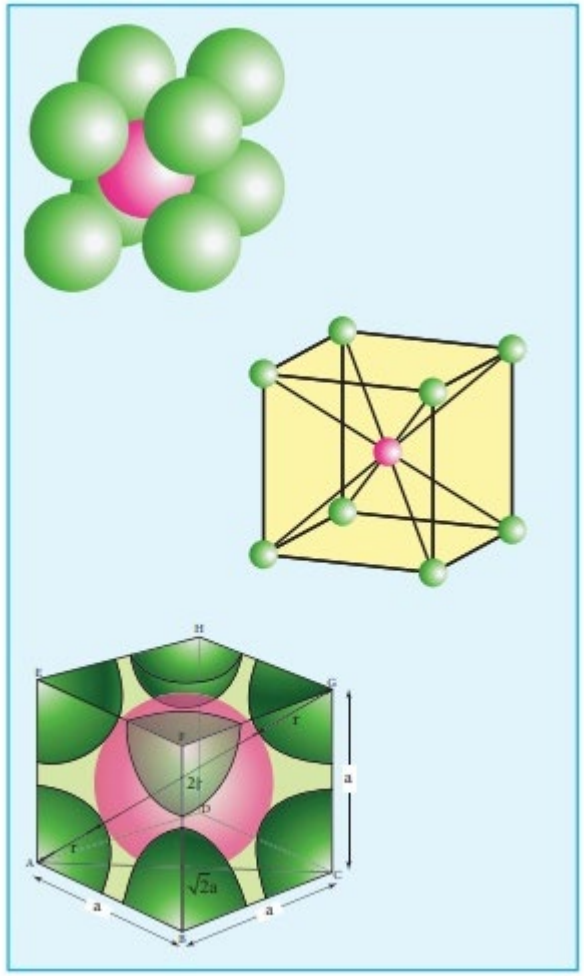
## Number of atoms in a unit cell:

### Simple Cubic (SC)



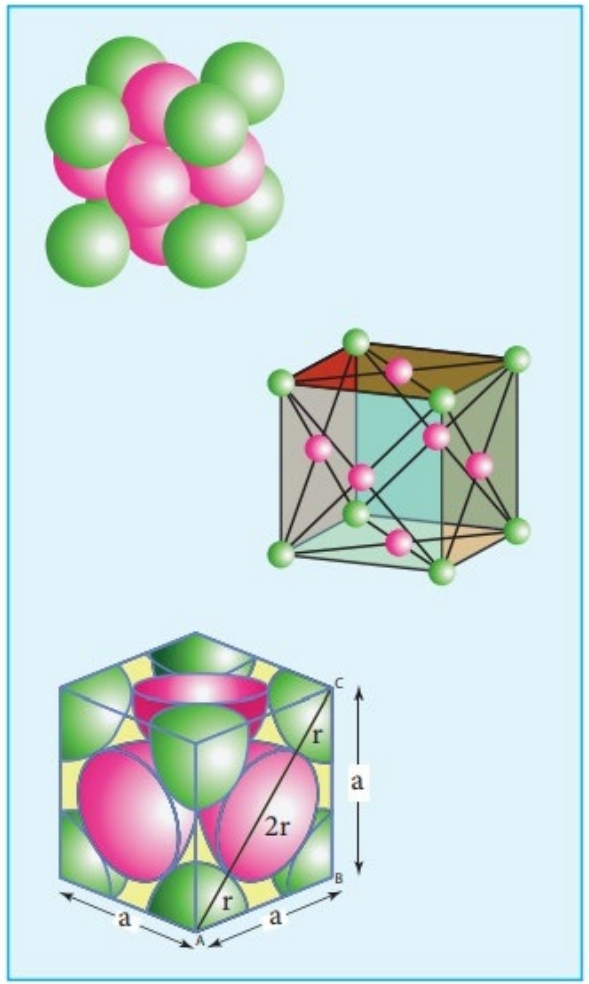

$N_a = 1$

### Body Centred Cubic (BCC)




**QUESTION:** $N_a = ?$

### Face Centred Cubic (FCC)




$N_a = ?$

## Crystal directions

Directions in lattices are determined relative to the crystal axes defined by the unit vectors of the unit cell.

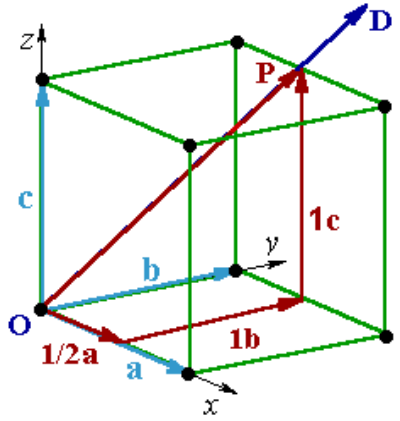
### Indexing of crystal directions

A particular direction index is expressed using square brackets [ ]
Directions in crystals are vectors which may be expressed in terms of the unit vectors **a**, **b**, and **c**:

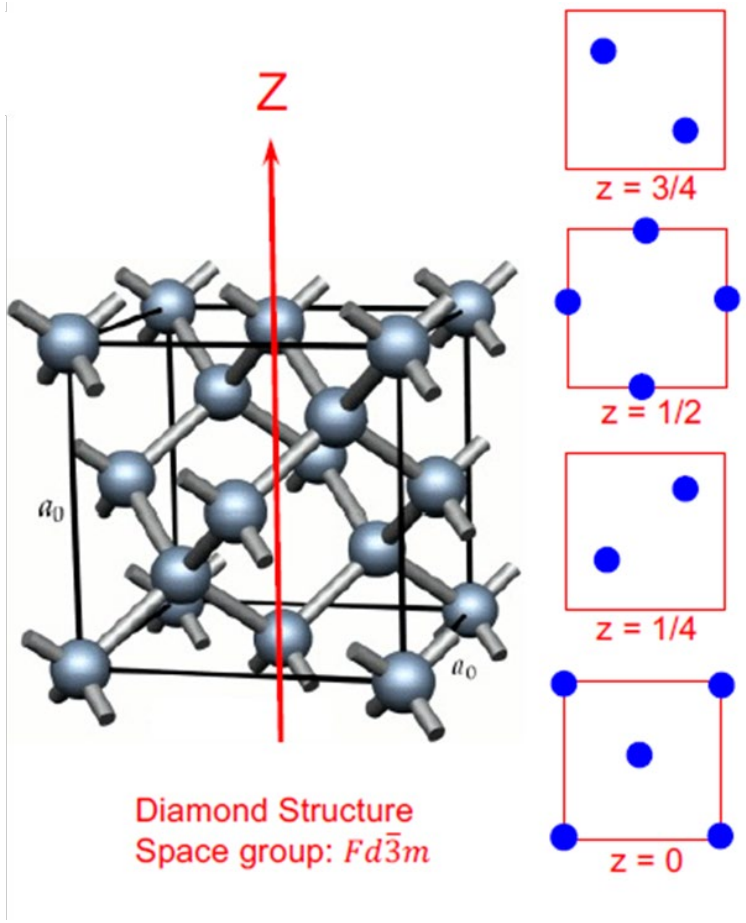
$$\mathbf{r}_{uvw} = u\,\mathbf{a} + v\,\mathbf{b} + w\,\mathbf{c}$$

In general, a **direction index** (or direction symbol, or **zone axis**) is given as **[u v w]** where u, v and w are *integers*.

If negative integers are required the notation is [-u -v -w], read as bar u, bar v, and bar w.



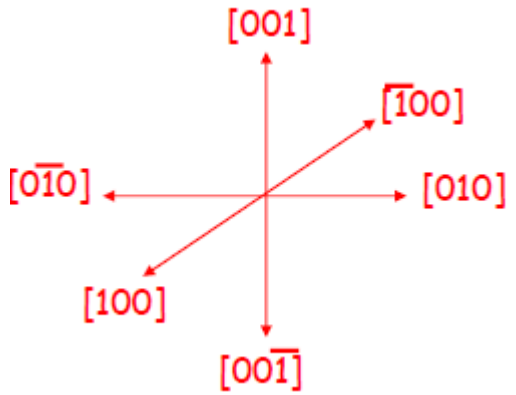

| In vector notation: | Direction of |
|---|---|
| OP = 1/2a + 1b + 1c | OD is [122] |

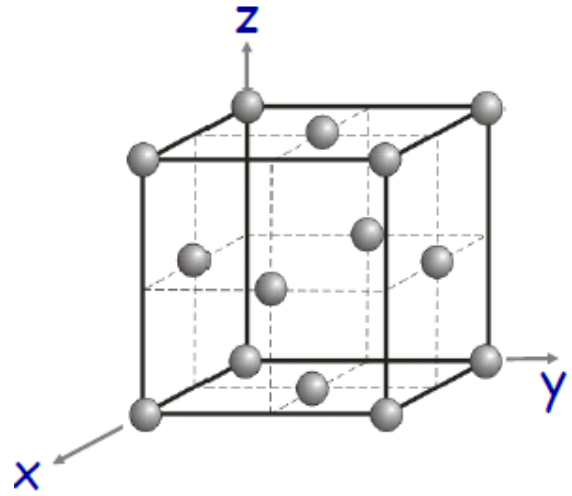


## Equivalent crystal directions

- Crystal directions [ ] and their families < >,

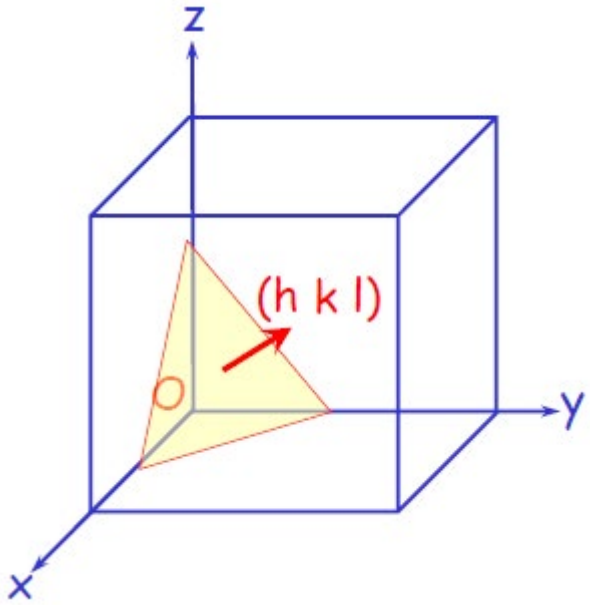
  as well as their determination.



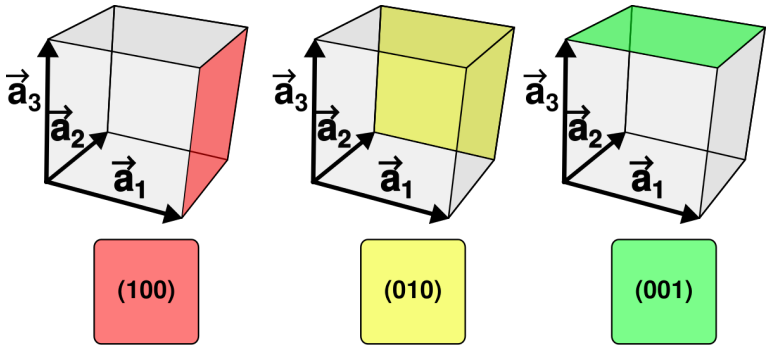

## Crystal planes via Miller indices

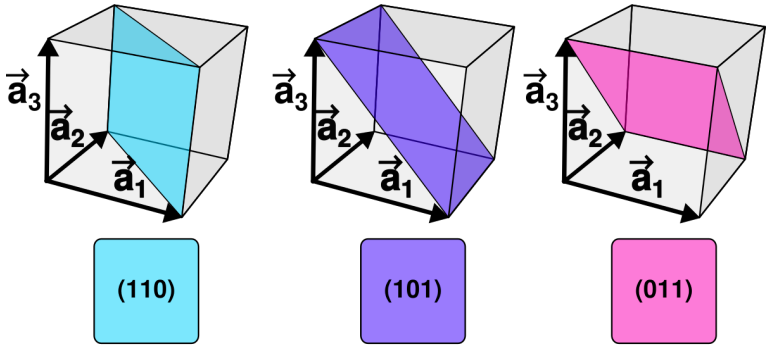
- Since crystal planes are two dimensional objects, the normal to a plane is used to define that plane.
- Miller indices are used to defined the crystal planes.
- A set of crystal planes form an infinite regularly spaced atomic planes, which collectively pass through all the points in a lattice.
- A set of crystal planes (parallel and equally spaced) has the same Miller index.
- Miller index ( ) of crystal planes and their families { }, as well as their determination.

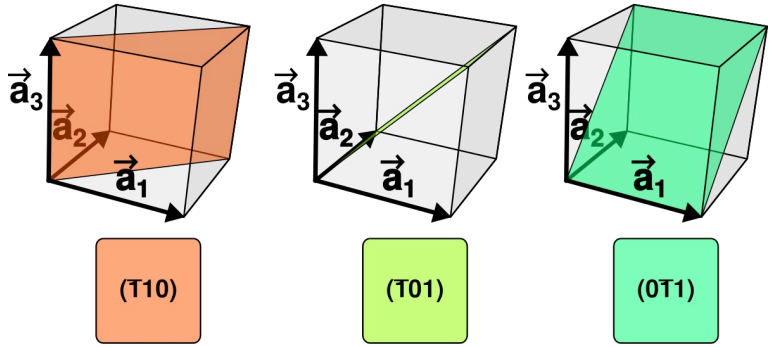


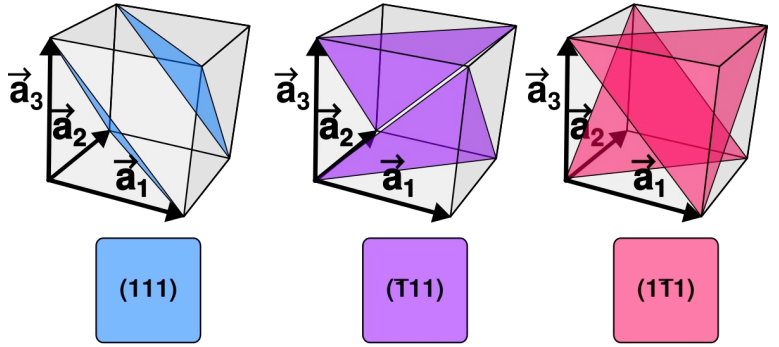
**Miller indexing of crystal planes:**

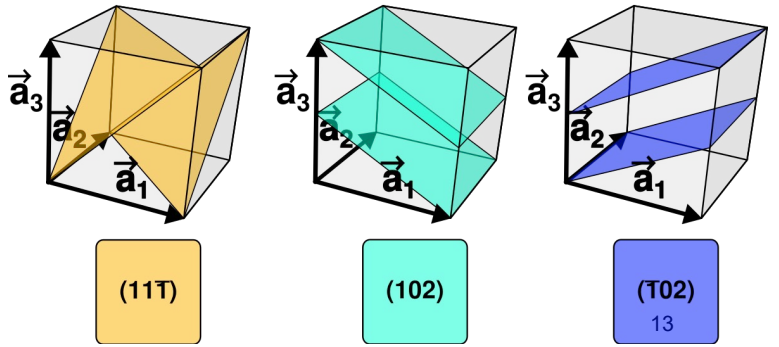
- A particular Miller index is expressed using round brackets ( ).
- A Miller index for a set of planes is expressed as (h k l), where h, k,& l are integers.
- If a crystal plane is parallel to any axis, its intercept on that axis is at infinity, the index for this axis is zero.
- If negative integers are required the notation is $(\bar{h}\ \bar{k}\ \bar{l})$, read as bar h, bar k, & bar l.



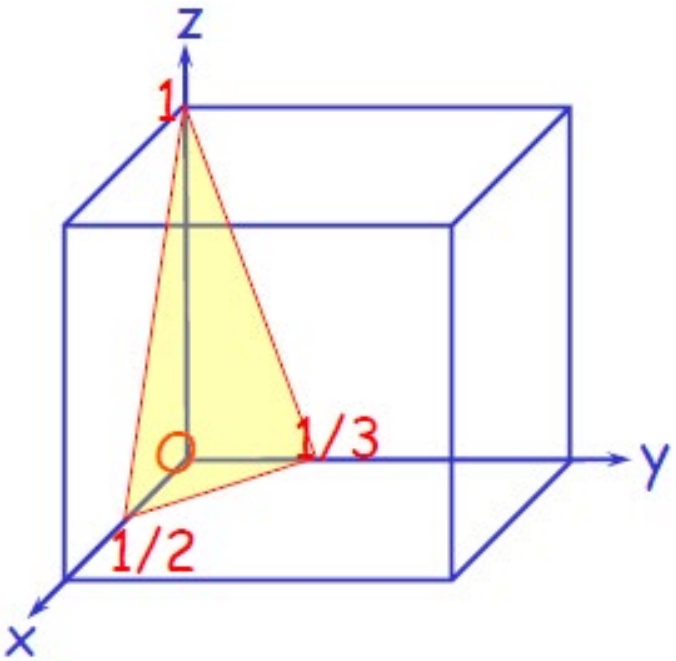



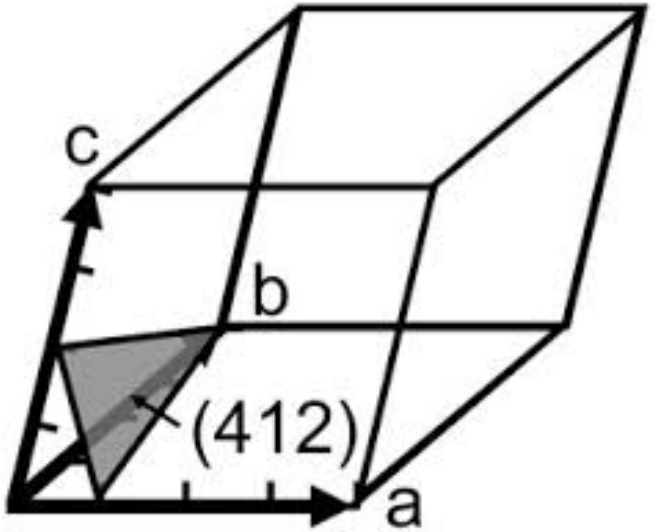




http://calistry.org/calculate/latticePlanesMillerIndices

## Determination of Miller index

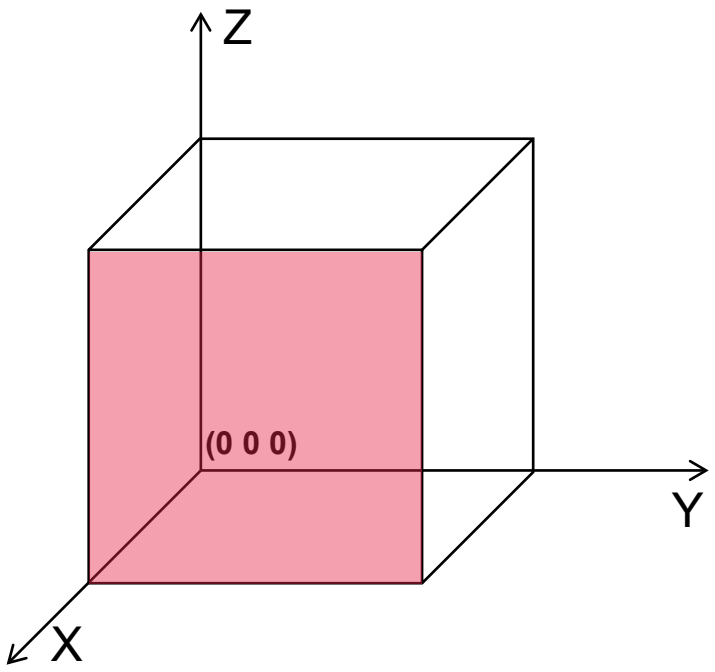
- Determine a set of parallel and equally spaced crystal planes to be indexed.
- Randomly choice the origin in one of these planes;
- Determine the intercepts (1/h)a, (1/k)b, (1/l)c on the axes for the first plane away from the origin.
- For example, (1/2)a, (1/3)b, (1/1)c. So, the plane is (2 3 1).
- If the plane is parallel to an axis, the intercept is ∞, the index for this axis is zero.



The plane which passes through crystal lattice and intersects lattice points is called “Lattice plane”.

Miller indices are set of three numbers used to specify lattice plane.




*Miller indices of this plane is (1 0 0)*

**Step 1:** Identify intercepts made by the plane in X, Y, Z axis, respectively.

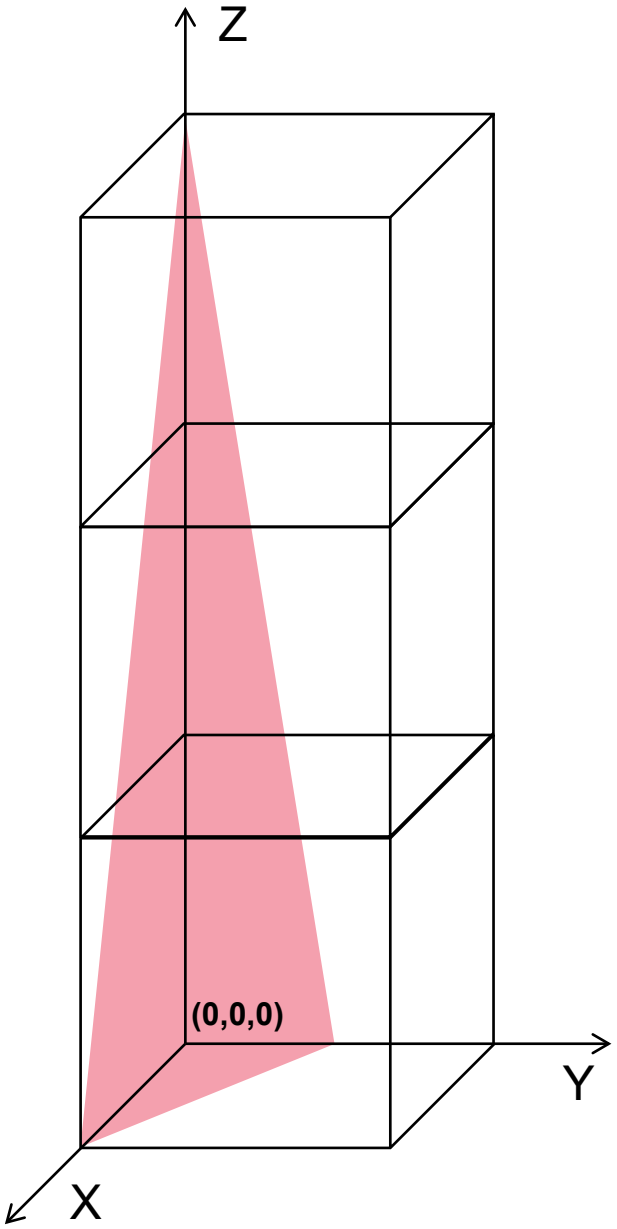
X intercept = 1
Y intercept = ∞
Z intercept = ∞

**Step 2:** Take reciprocal of each intercept.

X intercept = 1/1 = 1
Y intercept = 1/∞ = 0
Z intercept = 1/∞ = 0

**Rules to denote Miller indices:**

1. There is **no commas between numbers**.
2. The number of **Miller indices cannot be in the form of fraction**.
   *e.g.*, (0.66, 1, 0) or (2/3 1 0) the correct way is (2 3 0).




**Step 1:** Identify intercepts made by the plane in X, Y, Z axis, respectively.

X intercept = 1
Y intercept = 1/2
Z intercept = 3

**Step 2:** Take reciprocal of each intercept.

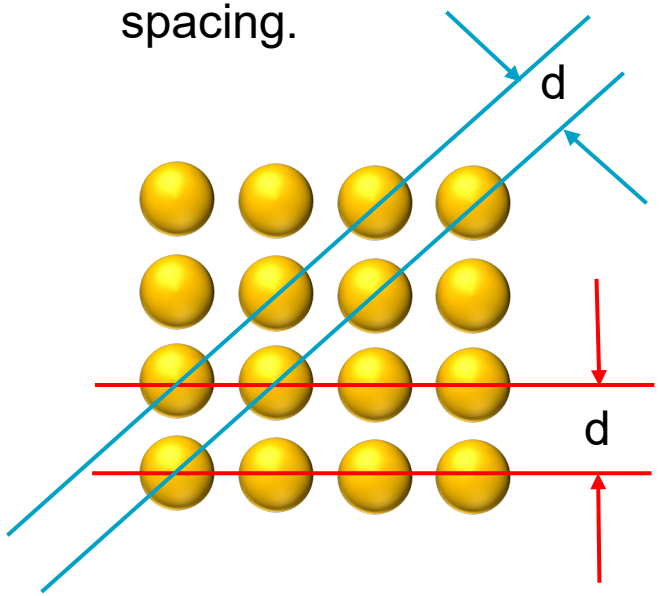
X intercept = 1/1 = 1
Y intercept = 1/0.5 = 2
Z intercept = 1/3

Miller indices of this plane is **(1 2 $^1/_3$)??**

No, the number cannot be in the form of fraction.
So by simply multiplying all numbers by 3, we can get Miller indices of this plane as (3 6 1).

## d-spacing of crystal planes

- The **spacing** of a set of planes (hkl), is the perpendicular distance between adjacent planes.
- The spacing is a function of (hkl).
- Generally, the higher the indices the smaller the spacing.

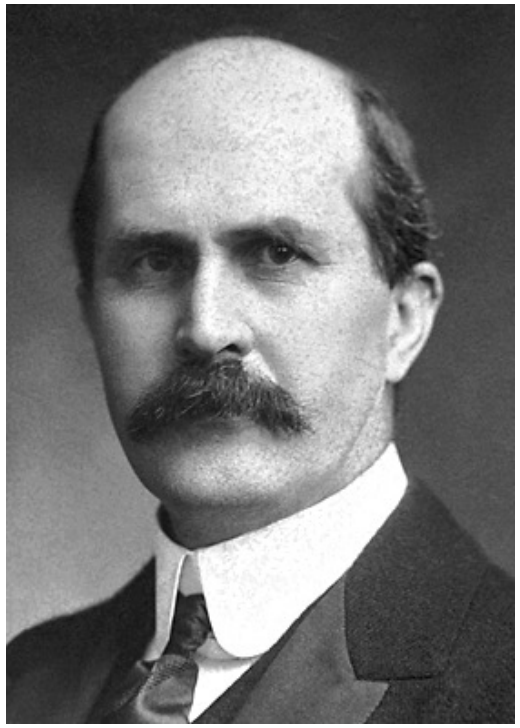


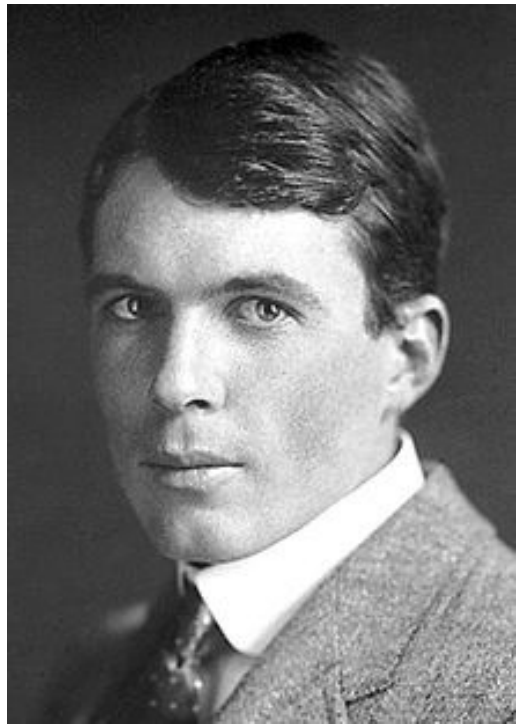

| Crystal System | Axial Translations | Axial Angles | Spacing, expressed as: $\frac{1}{d^2}$ |
|---|---|---|---|
| Cubic<br>P, F, I | a = b = c | $\alpha = \beta = \gamma = 90°$ | $\frac{h^2 + k^2 + \ell^2}{a^2}$ |
| Tetragonal<br>P, I | a = b ≠ c | $\alpha = \beta = \gamma = 90°$ | $\frac{h^2 + k^2}{a^2} + \frac{\ell^2}{c^2}$ |
| Hexagonal<br>P | a = b ≠ c | $\alpha = \beta = 90°, \gamma = 120°$ | $\frac{4}{3} \cdot \left(\frac{h^2 + hk + k^2}{a^2}\right) + \frac{\ell^2}{c^2}$ |
| Orthorhombic<br>P, I, C, F | a ≠ b ≠ c | $\alpha = \beta = \gamma = 90°$ | $\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{\ell^2}{c^2}$ |
| Rhombohedral<br>P | a = b = c | $\alpha = \beta = \gamma \neq 90°$ | $\frac{\left(h^2 + k^2 + \ell^2\right)\sin^2\alpha + 2 \cdot \left(hk + k\ell + h\ell\right)\cos^2\alpha - \cos\alpha}{a^2 \cdot \left(1 - 3\cos^2\alpha + 2\cos^3\alpha\right)}$ |
| Monoclinic<br>P, C | a ≠ b ≠ c | $\alpha = \gamma = 90°, \beta > 90°$ | $\frac{1}{\sin^2\beta} \cdot \left(\frac{h^2}{a^2} + \frac{k^2 \sin^2\beta}{b^2} + \frac{\ell^2}{c^2} - \frac{2h\ell \cdot \cos\beta}{ac}\right)$ |
| Triclinic<br>P | a ≠ b ≠ c | $\alpha \neq \beta \neq \gamma \neq 90°$ | $\frac{1}{V^2} \cdot \left(S_{11} \cdot h^2 + S_{22} \cdot k^2 + S_{33} \cdot \ell^2 + 2S_{12} \cdot hk + 2S_{23} \cdot k\ell + 2S_{13} \cdot h\ell\right)$<br>V = volume of unit cell<br>$S_{11} = b^2c^2 \sin^2\alpha$<br>$S_{22} = a^2c^2 \sin^2\beta$<br>$S_{33} = a^2b^2 \sin^2\gamma$<br>$S_{12} = abc^2 \cdot \left(\cos\alpha\cos\beta - \cos\gamma\right)$<br>$S_{23} = a^2bc \cdot \left(\cos\beta\cos\gamma - \cos\alpha\right)$<br>$S_{13} = ab^2c \cdot \left(\cos\gamma\cos\alpha - \cos\beta\right)$ |

## Pioneers of diffractions



Max von Laue (1879-1960) won the Noble Prize in Physics, 1914 *for his discovery of the diffraction of X-ray by crystals*!

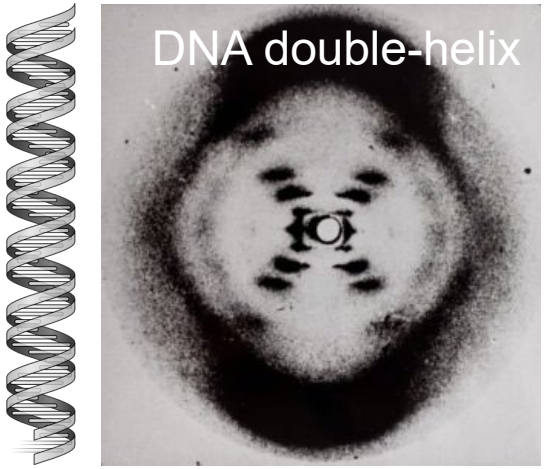


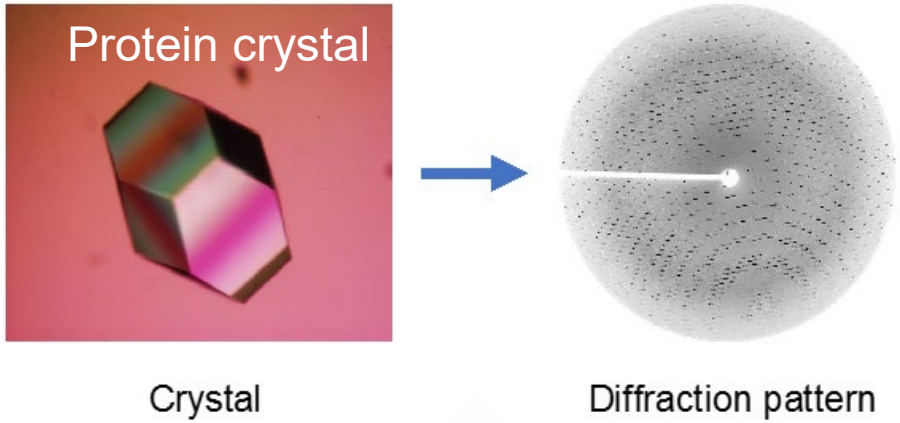


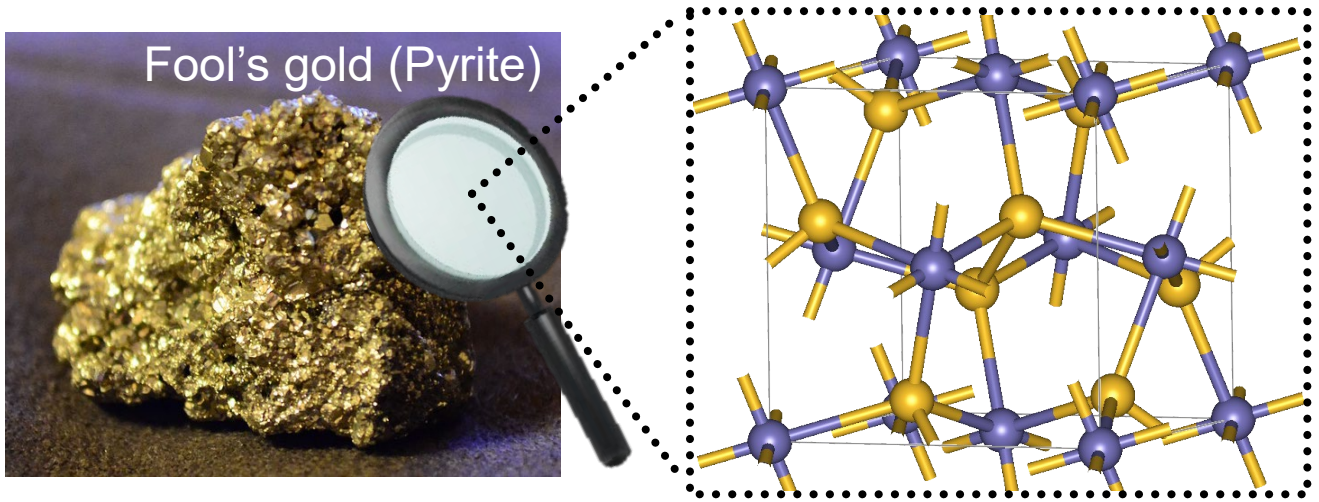
William Henry Bragg (1862-1942) and his Australian born son William Lawrence Bragg (1890-1971) won the Noble Prize in Physics, 1915 *for their services in the analysis of crystal structure by means of X-ray*.

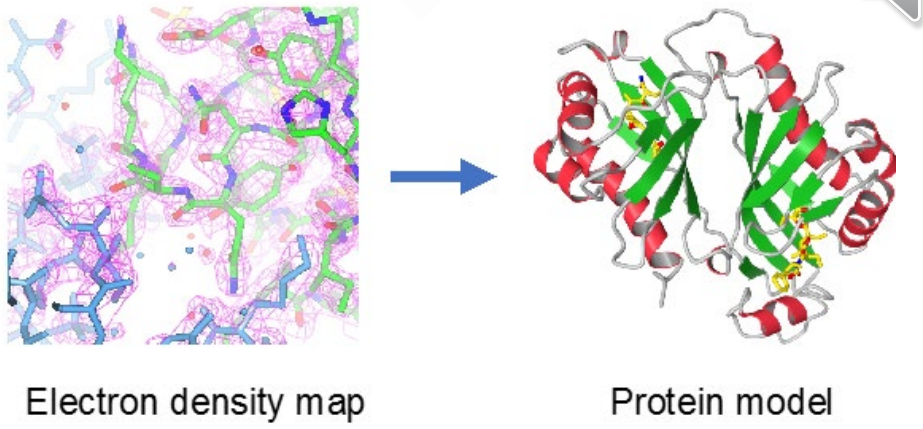
The finding of X-ray revolutionized various fields including
Medical imaging, genetics, proteomics, materials science, manufacturing…



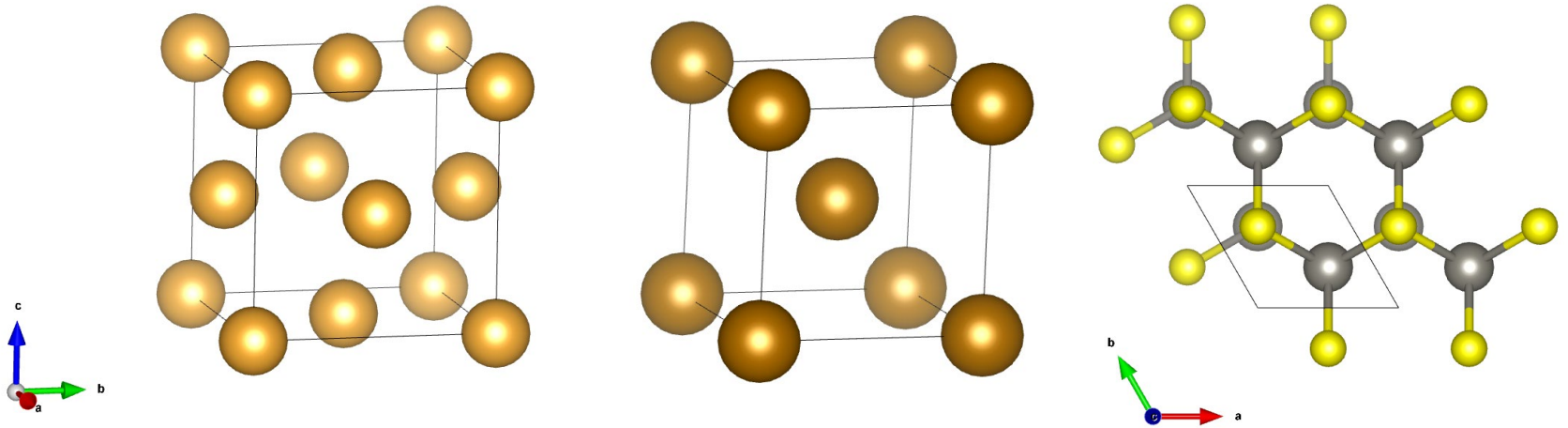



## What is XRD?

X-ray diffraction (XRD) is a technique by irradiating a material with incident X-rays and measuring the intensities and scattering angles of the X-rays that leave the material.




**Risk Involved:**

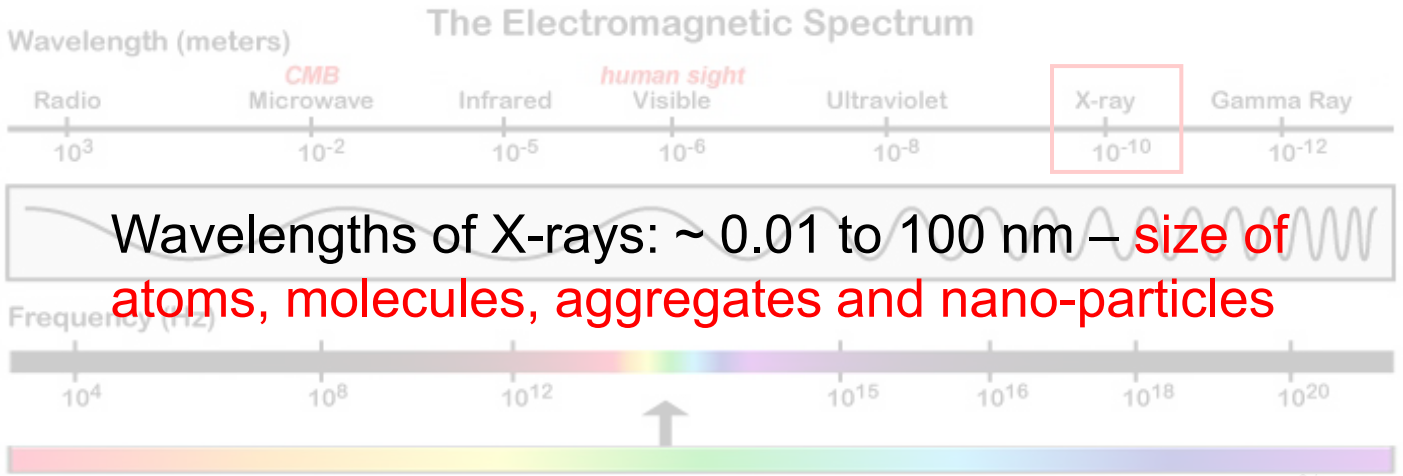



Radiation Safety

## What material information can be obtained?

It is used to identify

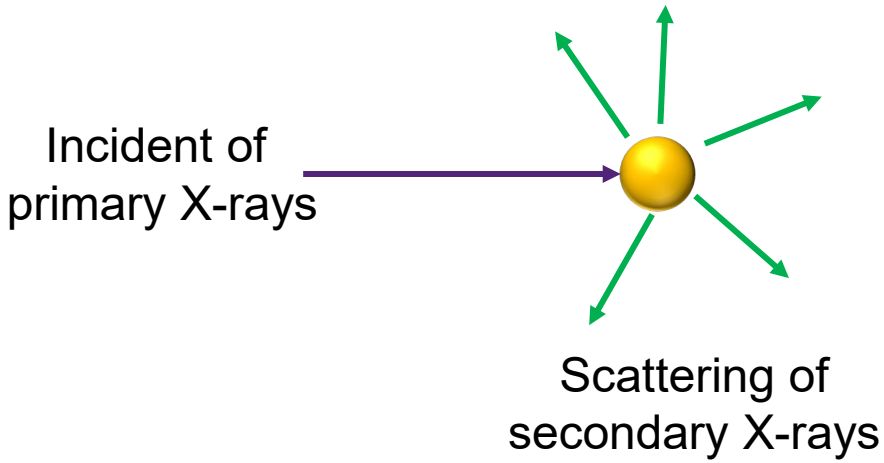
- Crystalline phases and orientation
- Determine structural properties:
  - Lattice parameters
  - Crystallite size
  - Phase composition
  - Strain




Wavelengths of X-rays: ~ 0.01 to 100 nm – size of atoms, molecules, aggregates and nano-particles

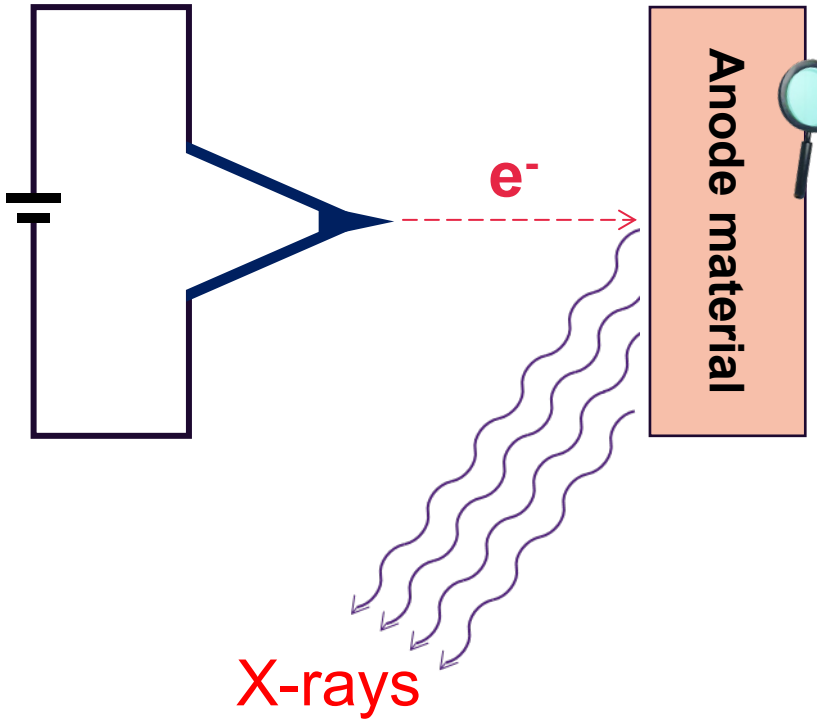
## Scattering and diffraction

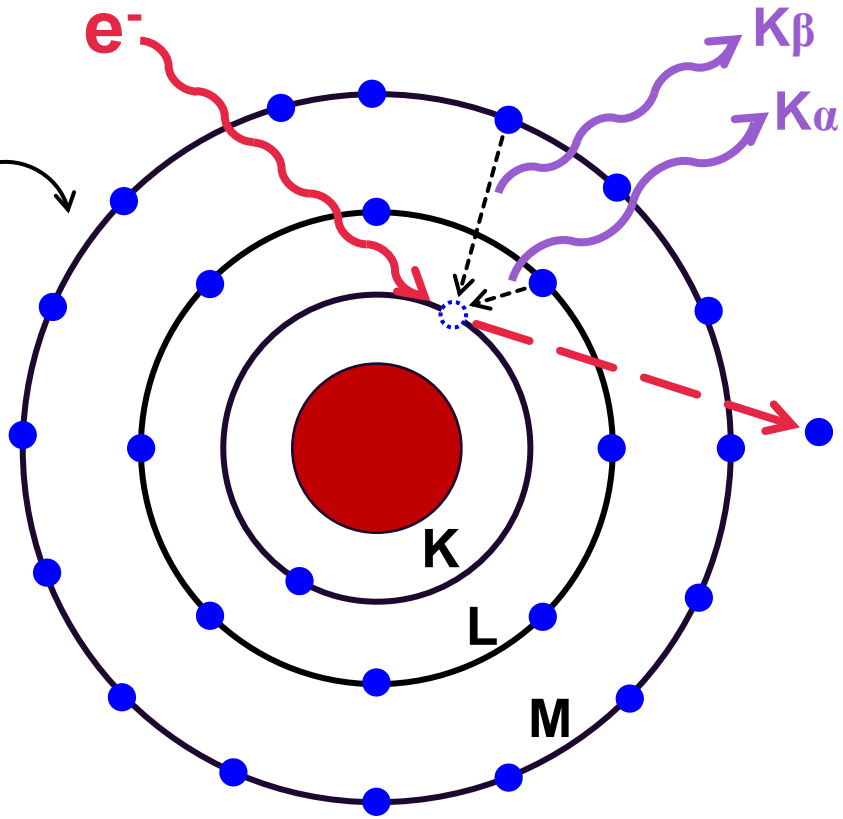
**Diffraction** is the **sum of scattering** events! When an incident beam (radiation) hits a particle, the wave interacts with a single particle, the particle scatters the incident beam uniformly in all directions.



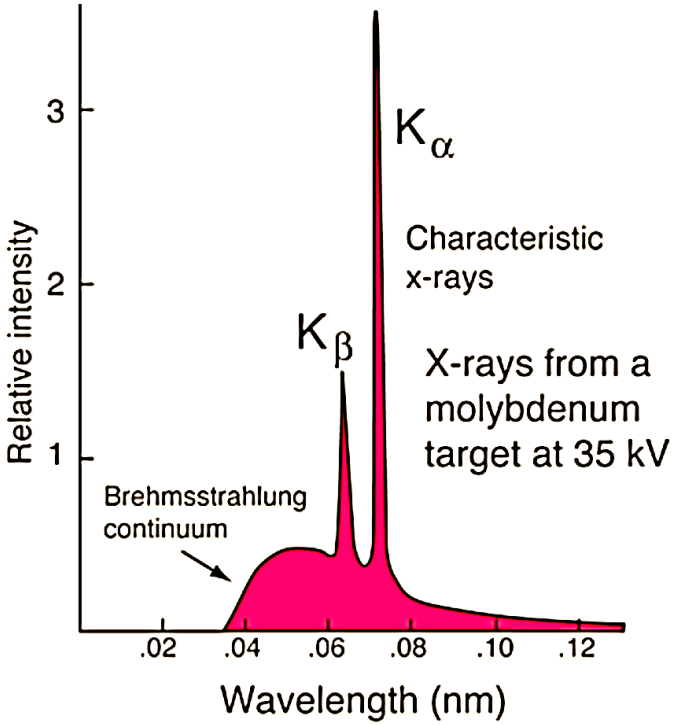

- Diffraction techniques exploit the scattering of radiation from large numbers of sites.
  - Scattering from atoms
  - Scattering from groups of atoms in crystals
- Diffraction is an interference effect which leads to scattering of strong
- beams of radiation in specific directions.
- There are various diffraction techniques: X-ray diffraction, Electron diffraction, Neutron diffraction

## Generation of X-rays




## Characteristic X-rays (Incident X-ray for XRD)




Continuous X-ray / Characteristic X-rays

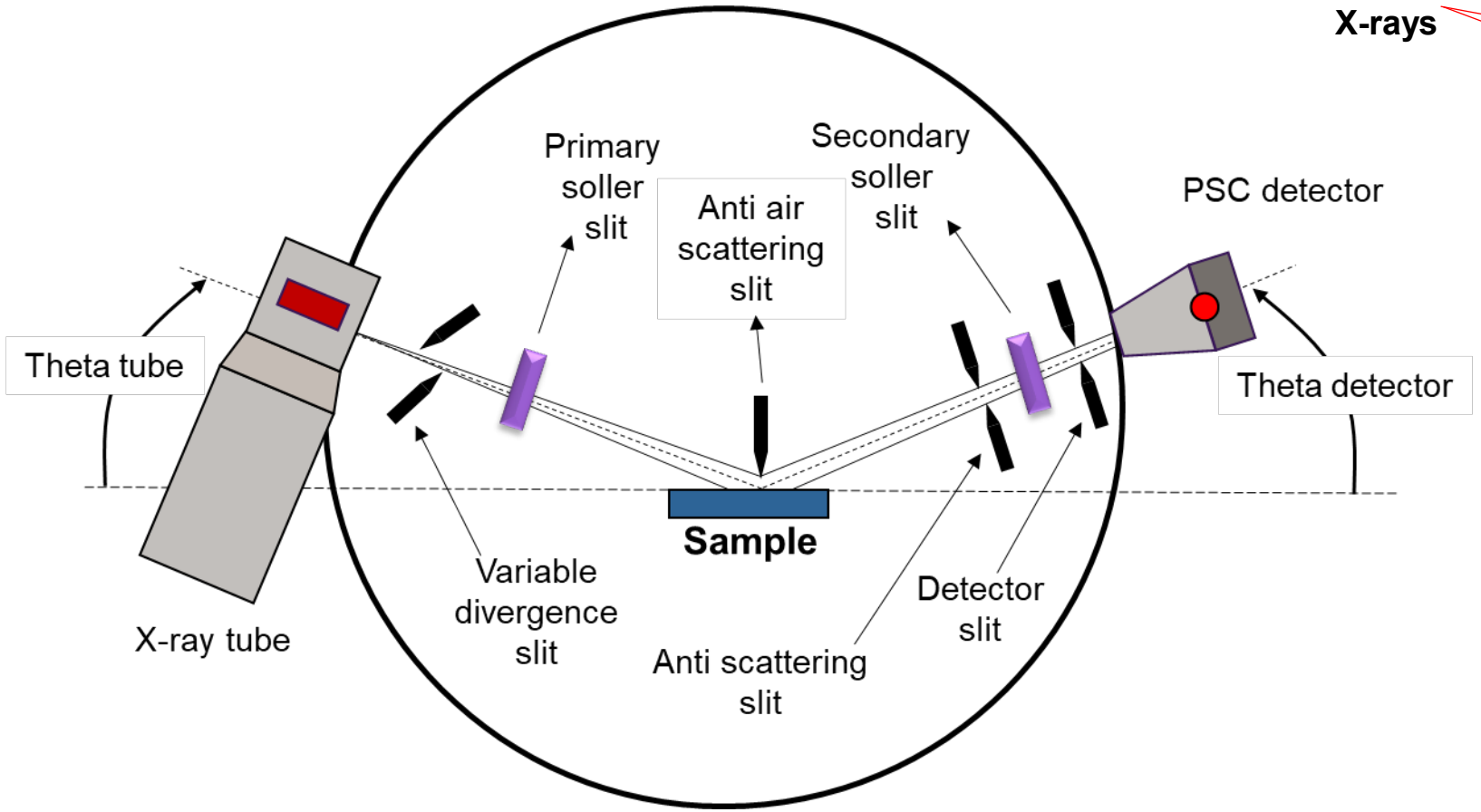
**Continuous X-rays**

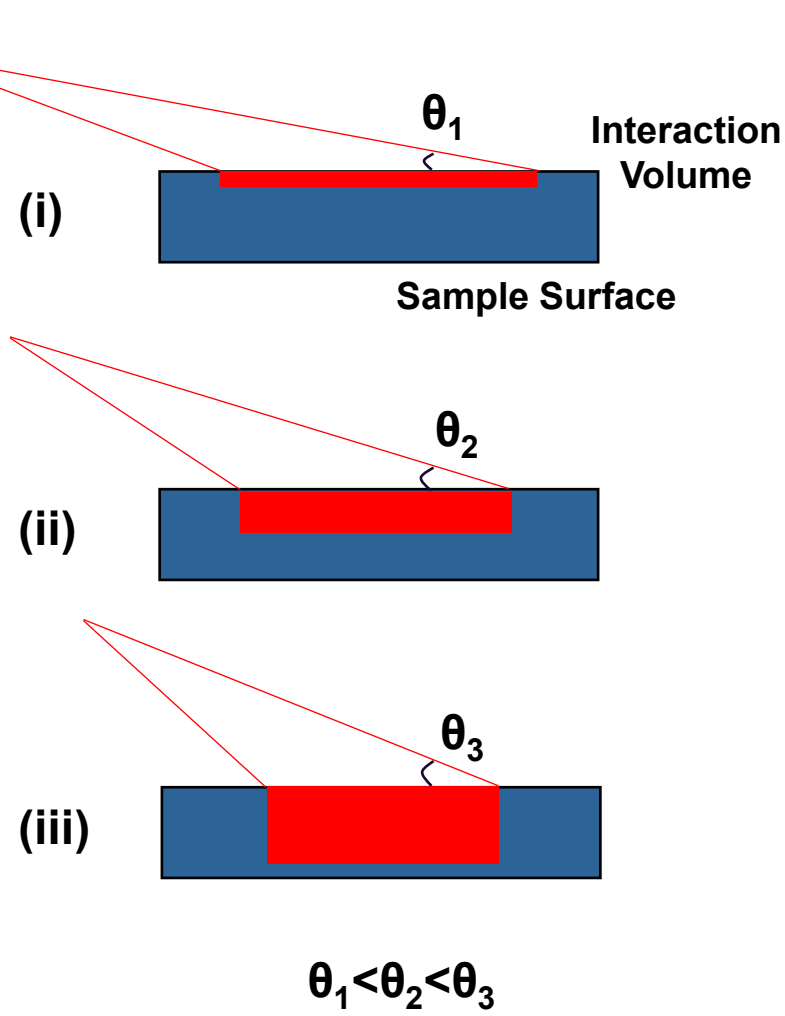
- radiation from braking

$$\lambda_{\min} = \frac{hc}{eV} = \frac{1.24}{V(kV)} nm$$

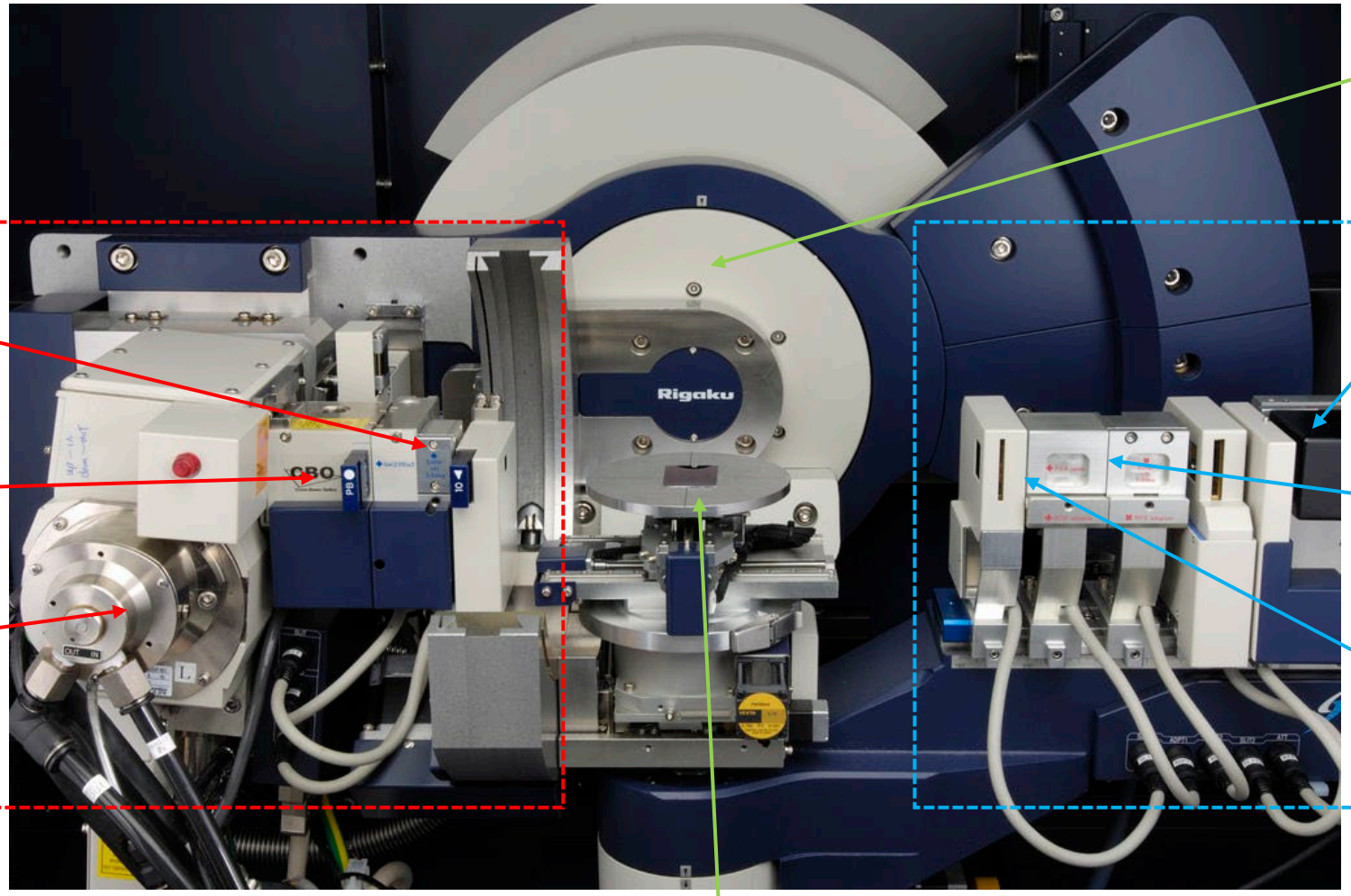
| Target | Cr | Fe | Co | Ni | Cu | Mo | Ag |
|---|---|---|---|---|---|---|---|
| Atom # | 24 | 26 | 27 | 28 | 29 | 42 | 47 |
| Kα (Å) | 2.291 | 2.937 | 1.790 | 1.659 | 1.541 | 0.711 | 0.561 |

## Fundamentals of XRD

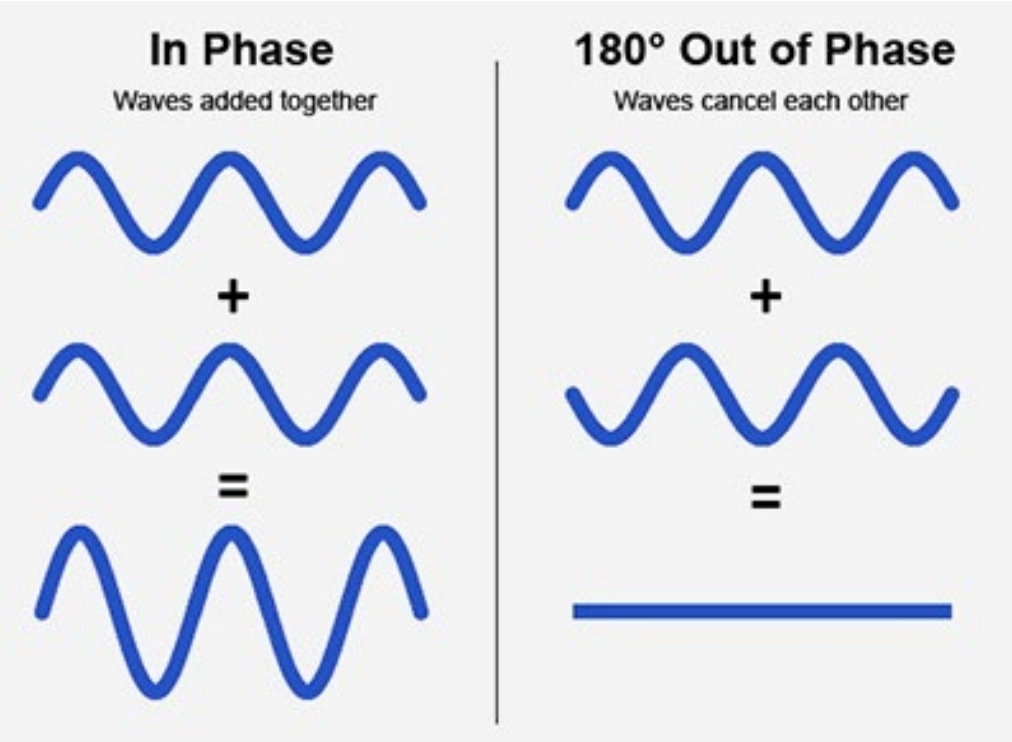


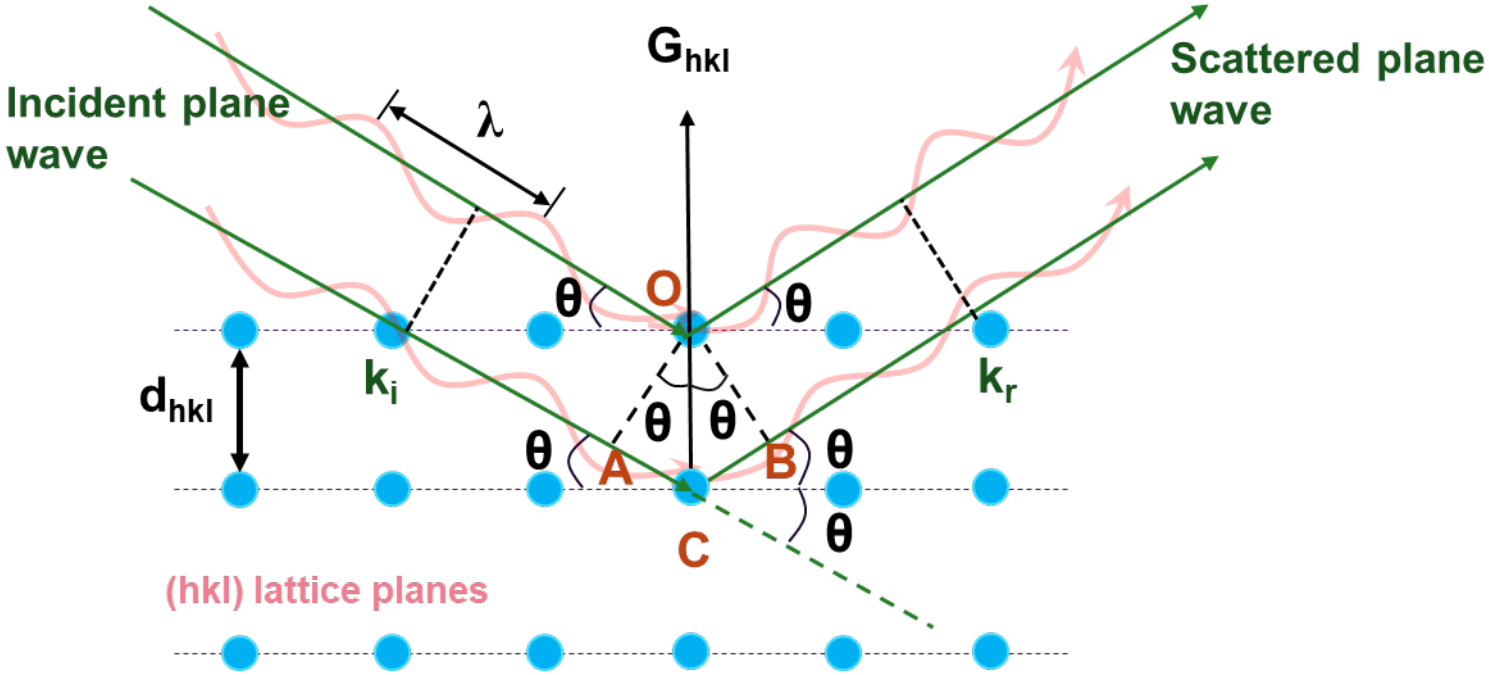




- Fundamental components in an X-ray diffractometer and its reflected angle (θ) dependent change in the X-ray interaction volume on sample surface.


Goniometer
SOURCE
DETECTOR
Divergence slit
Primary x-ray optics
X-ray tube
Detector
Soller slits
Diffraction beam collimator
Sample Stage
Rigaku

## Operating Principle



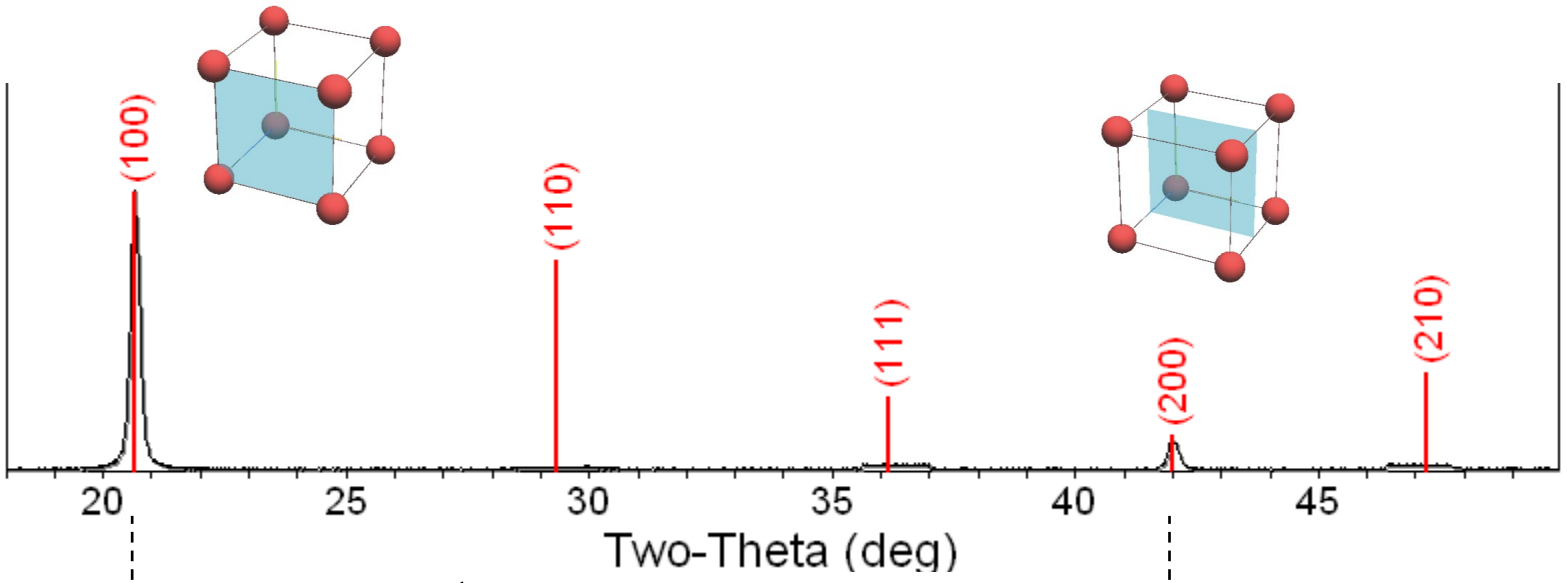

From ΔOCB,

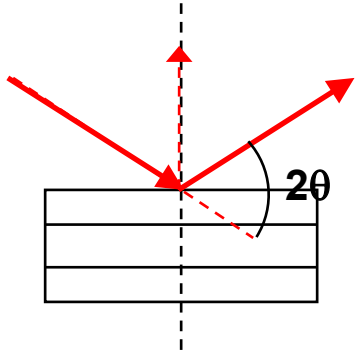
*BC = d sinθ* ; where, d = OC ( interplanar distance) and BC=AC

The total path difference between two x-rays that escape from the crystals is given by: *AC+BC = 2d sinθ*

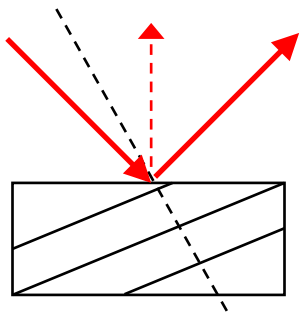
A constructive interference that results in the bright band (also called Bragg's peak), is observed when the path difference between diffracted X-rays is equal to, *nλ = 2d sinθ.*

## A single crystal specimen in a Bragg-Brentano diffractometer would produce only one family of peaks in the diffraction pattern.

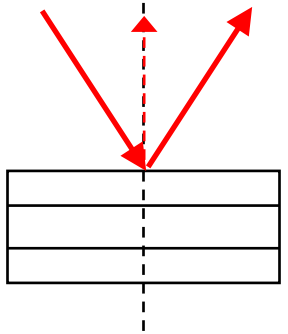



At 20.6 °2θ, Bragg's law fulfilled for the (100) planes, producing a diffraction peak.

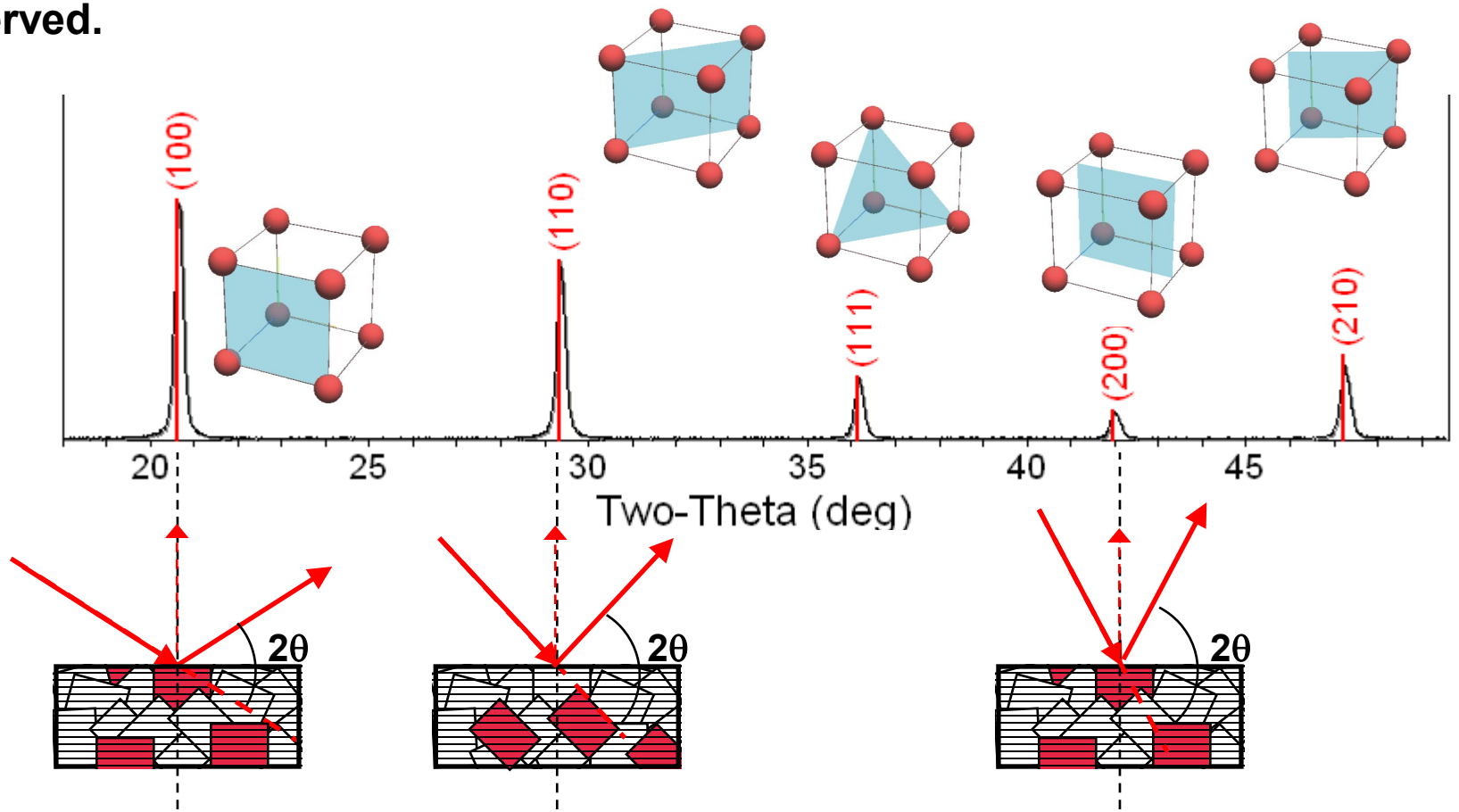


The (110) planes would diffract at 29.3 °2θ; however, they are not properly aligned to produce a diffraction peak (the perpendicular to those planes does not bisect the incident and diffracted beams). Only background is observed.



The (200) planes are parallel to the (100) planes. Therefore, they also diffract for this crystal. Since $d_{200}$ is ½ $d_{100}$, they appear at 42 °2θ.

**A polycrystalline sample should contain thousands of crystallites. Therefore, all possible diffraction peaks should be observed.**

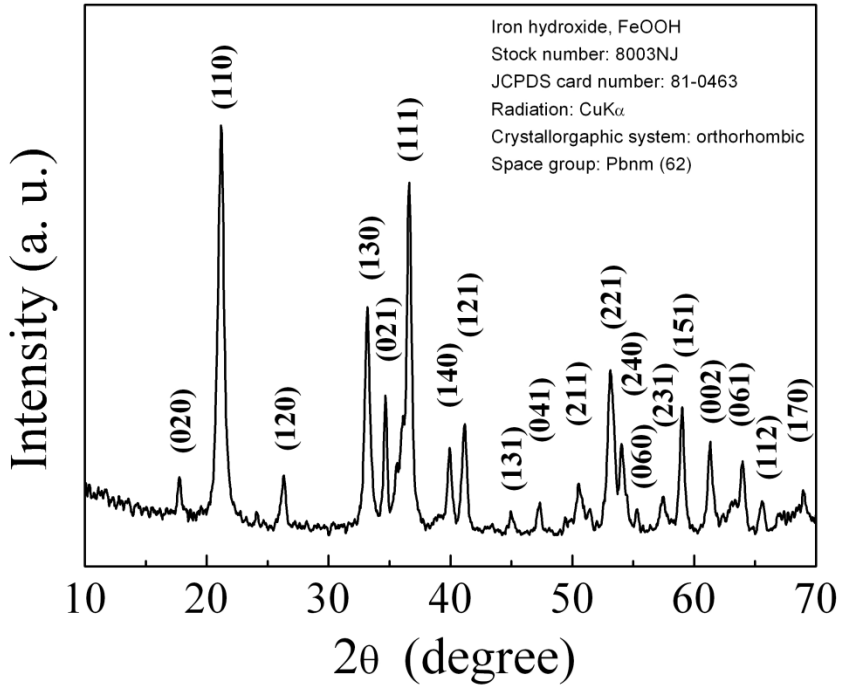



- For every set of planes, there will be a small percentage of crystallites that are properly oriented to diffract (the plane perpendicular bisects the incident and diffracted beams).
- Basic assumptions of powder diffraction are that for every set of planes there is an equal number of crystallites that will diffract and that there is a statistically relevant number of crystallites, not just one or two.

## Example: Diffraction patterns from XRD

Each peak represents the solution to Bragg's law for known (*i.e.*, Cu kα) radiation wavelength (λ = 0.15406nm).

The unique relationship between such patterns and crystal structures provide a powerful tool for identification of the phase composition of powders and polycrystalline materials.



# Relationship between diffraction peaks, miller indices and lattice spacings

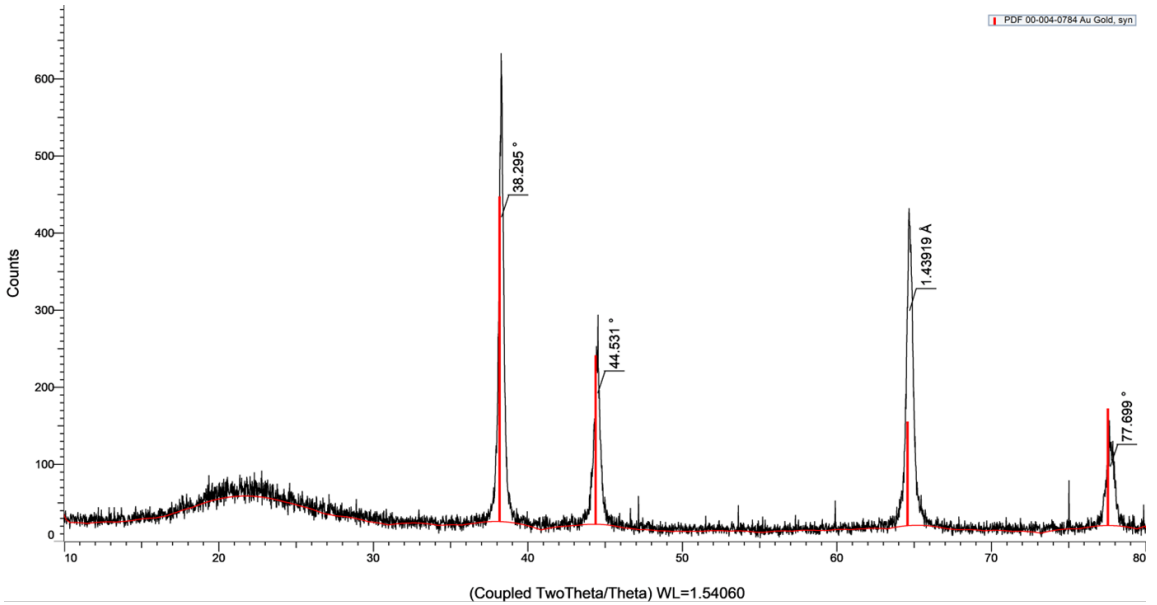
Simple cubic material **a** = 5.0 Å

| hkl | d(Å) | 2Θ |
|---|---|---|
| 100 | 5.00 | 17.72 |
| 110 | 3.54 | 25.15 |
| 111 | 2.89 | 30.94 |

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Bragg Equation: **nλ = 2dsinθ**

How many lattice planes are possible?
How many d-spacings? The number is large but finite.
**nλ = 2dsinθ** so if theta = 180, then d = λ/2. For Cu radiation that means that we can only see d-spacings down to 0.77 Å for Mo radiation, down to about 0.35 Å



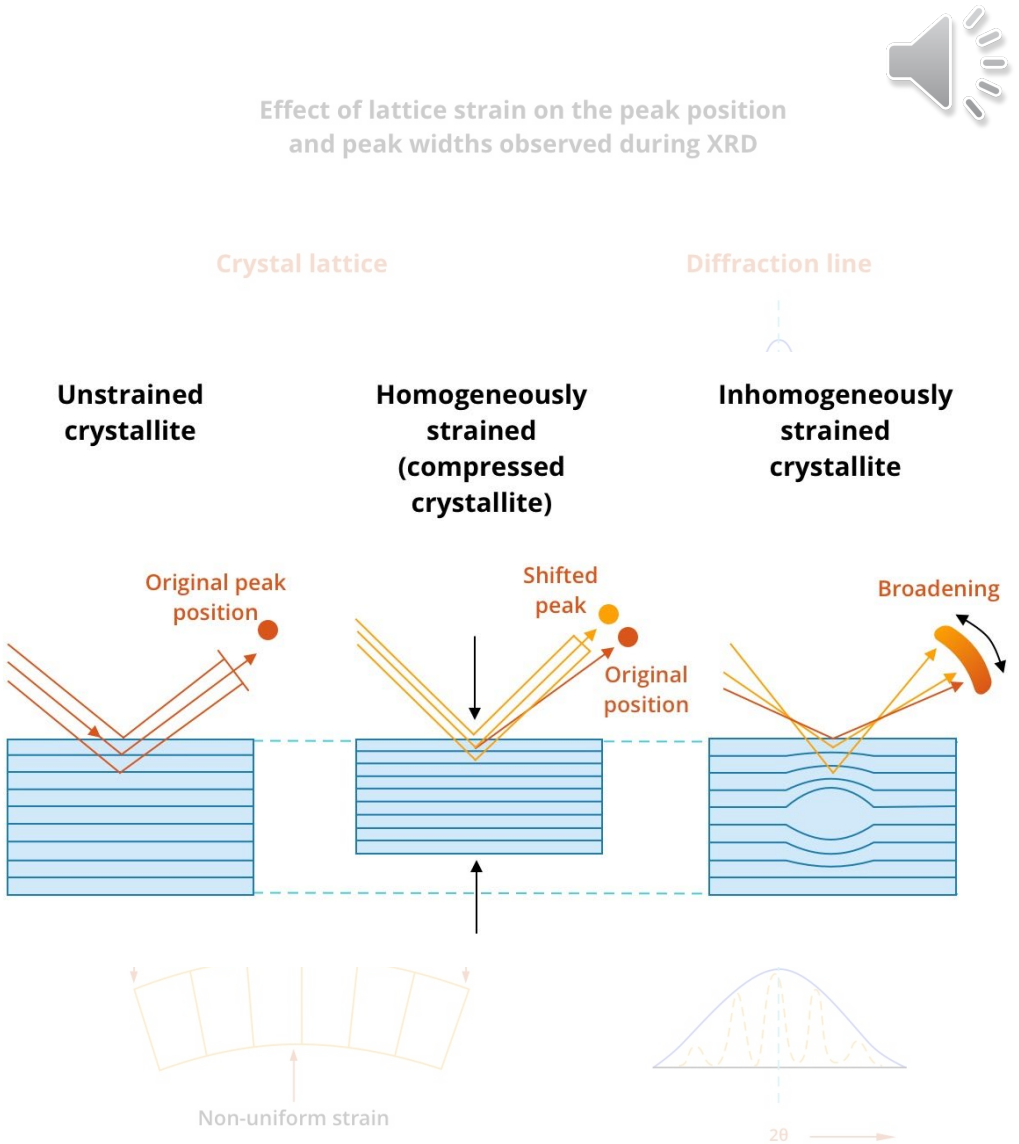

Phase identified (*PDF 00-004-0784, Au*)

$$\lambda = 2d_{hkl} \sin\theta_{hkl}$$

**Note:** Different anode gives different λ, so 2θ varies with the anode material.

# Basis and Bravais Structure Factor Terms



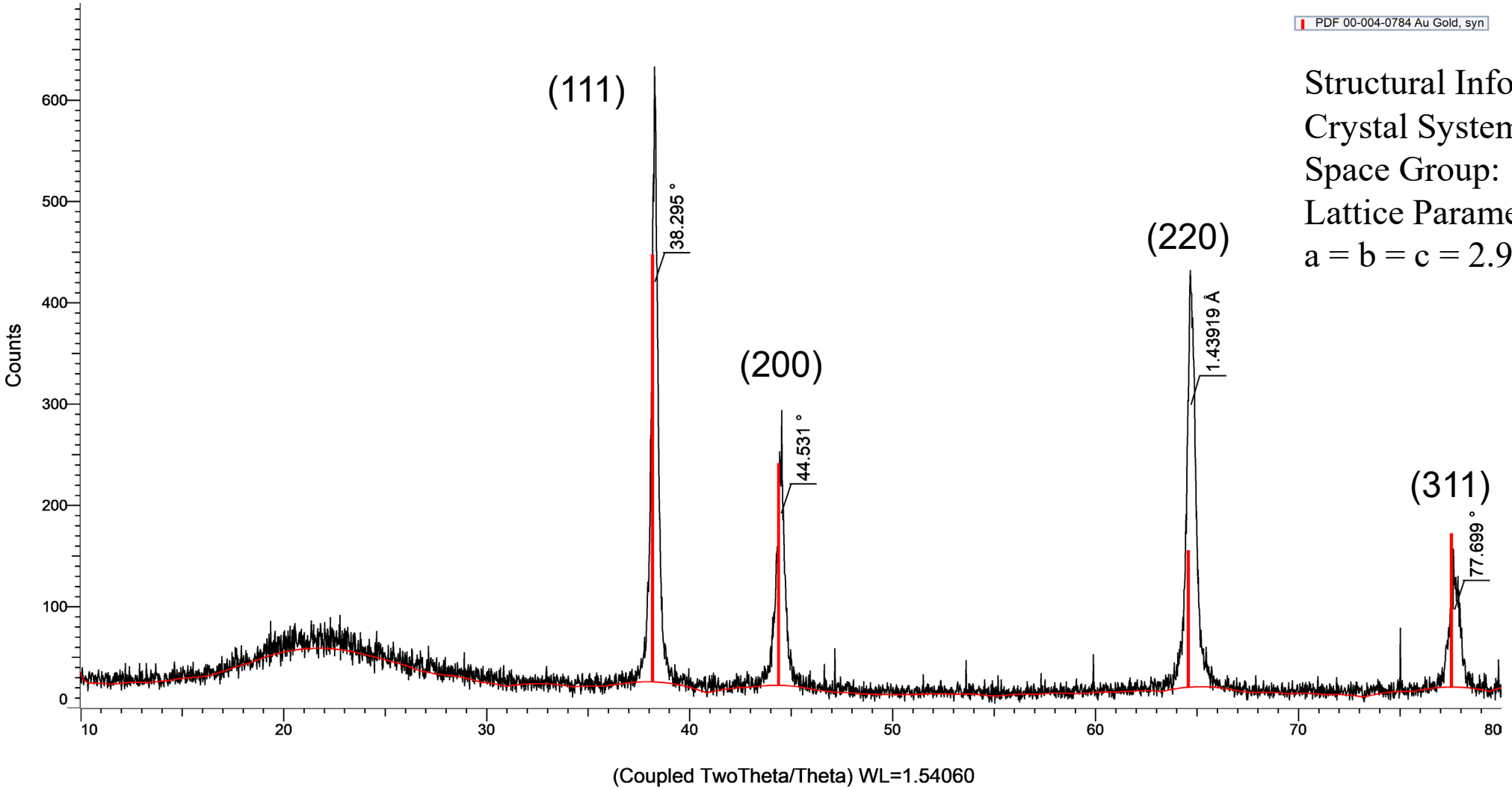



## What information we can get from an XRD analysis?

- Lattice parameters
- Phase identity
- Phase purity
- Crystallinity
- Crystal structure
- Percent phase composition

## What information we can not get from an XRD analysis?

- Elemental analysis -
  - How much lithium is in this sample?
  - Is there iron in this sample
  - What elements are in this sample
- Tell me what this sample is ????
  - Unless you know something about this sample, powder XRD won't have answers !!!

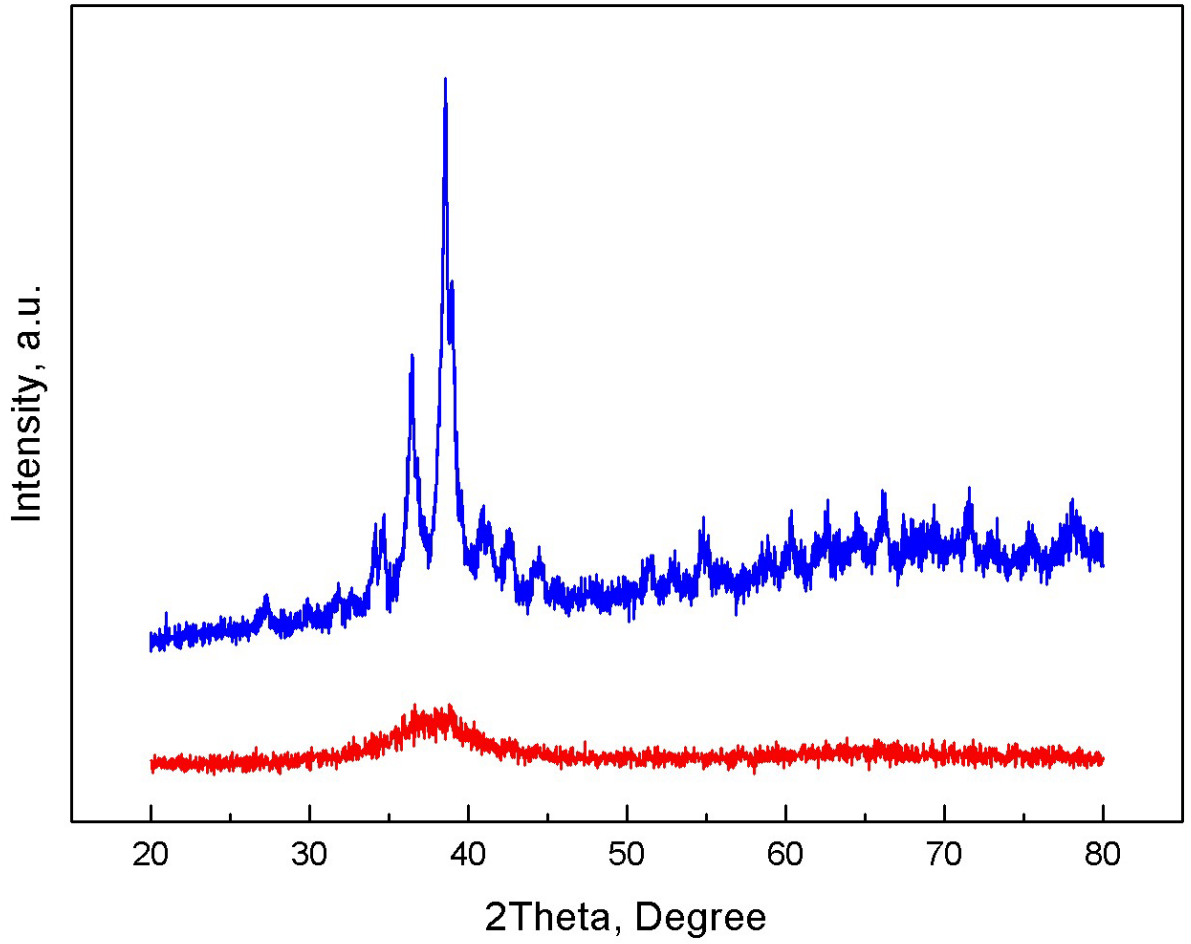



Structural Information: **Au**
Crystal System: Cubic
Space Group: $Fm\bar{3}m$
Lattice Parameters:
$a = b = c = 2.95Å$

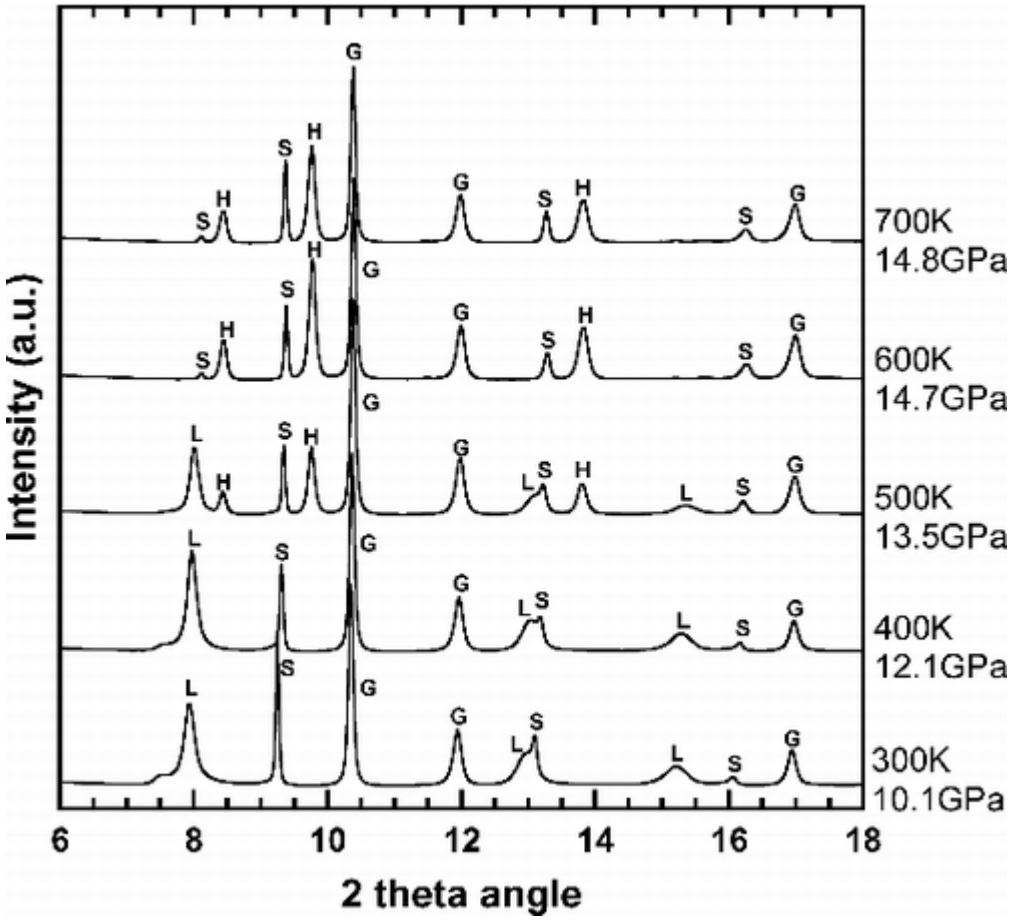
Phase identified (*PDF 00-004-0784, Au*)

## Examples of X-ray diffraction

Monitoring crystallization from amorphous




Monitoring phase change during materials processing



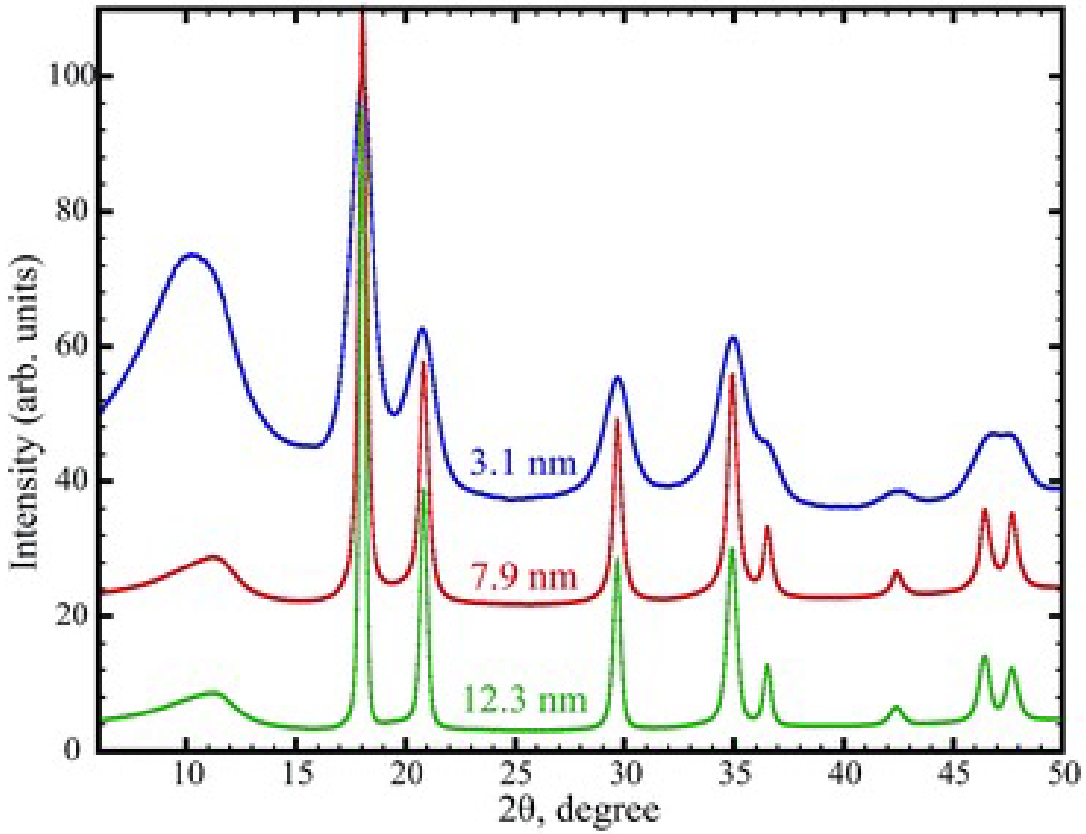
## X-ray diffraction of nanocrystals

- XRD patterns of Pt nanoparticles with different sizes.
- The **diffraction peaks become broadening with decreasing particle size**!
- The width of the diffraction peaks provides information on the persistence of long-range order and hence the crystal size.
- Practically, the particle size($D_p$) can be estimated by,

Debye-Scherrer equation:

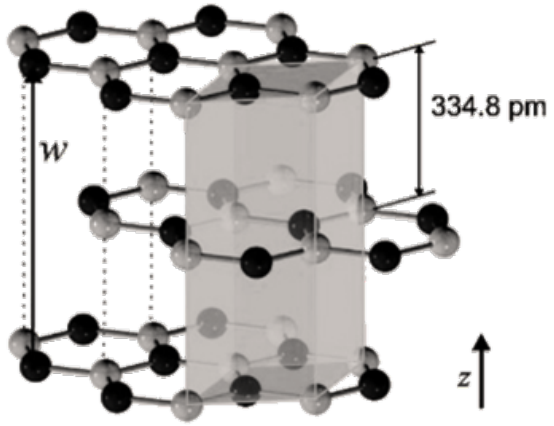
$$D_p = \frac{K\,\lambda}{\beta\,cos\theta}$$

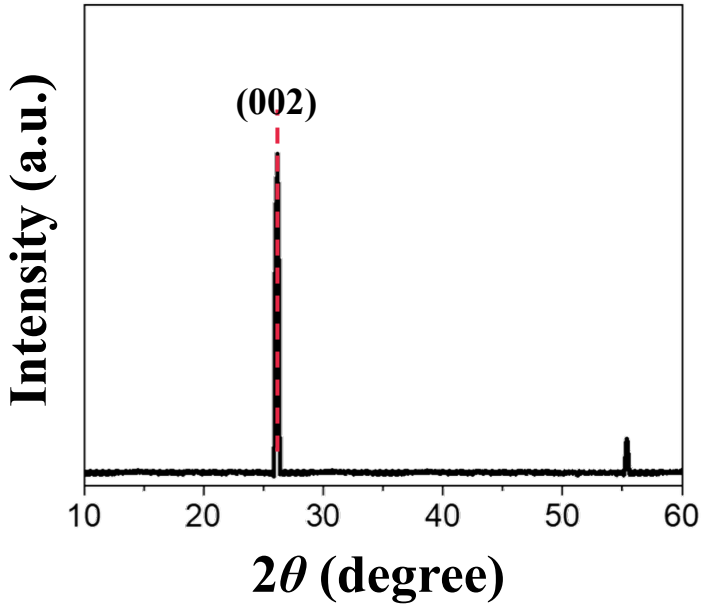
Where,
K = Scherrer's Constant (K=0.94)
β = Line broadening (in radians),
θ = Bragg angle (in radians),
λ = X-Ray wavelength (*i.e.,* for Cu kα 0.15406 nm )



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## Graphitised carbon (graphite)

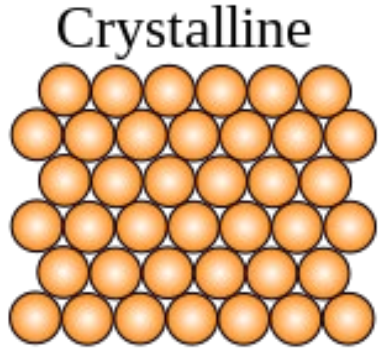


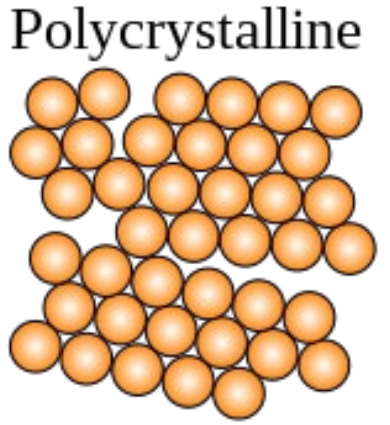




Crystalline material

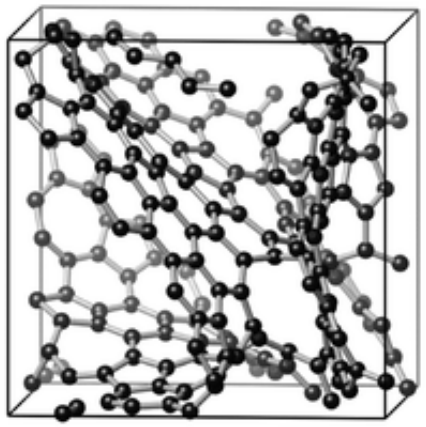
Uniform d-spacing of specific lattice planes.

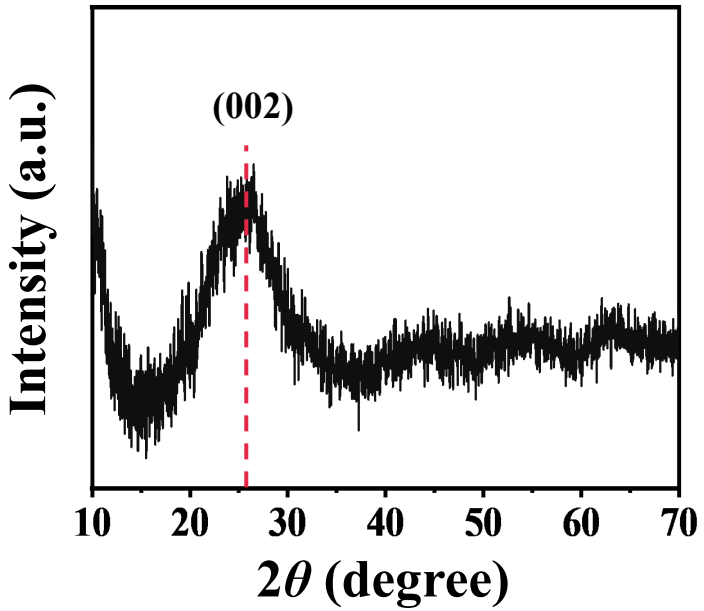
Sharp diffraction peaks at respective angles.




## Amorphous carbon

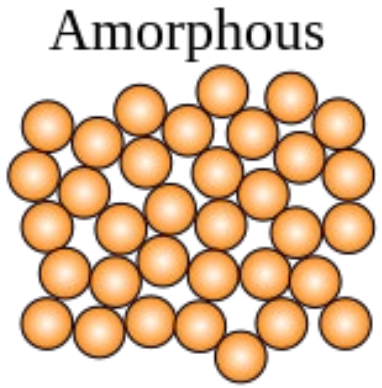





Amorphous material

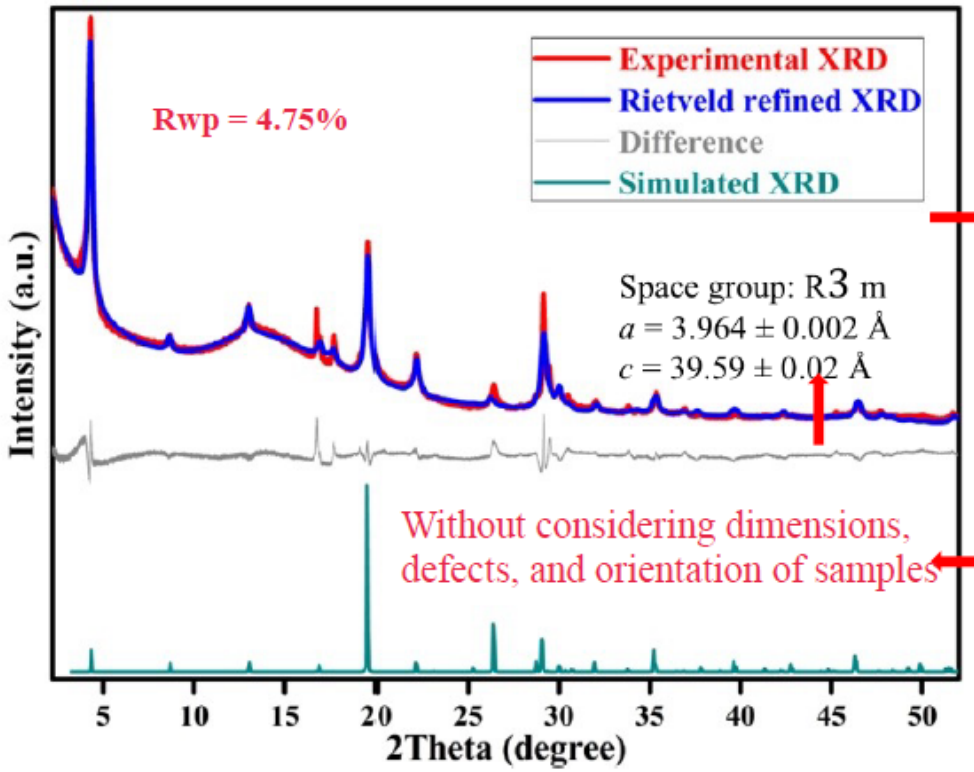
Random d-spacing of specific lattice planes.

Broad diffraction peaks at respective angles.



## Synchrotrons X-ray diffraction (XRD)

### Structure determination of newly discovered $In_3Se_4$ phase using synchrotron XRD

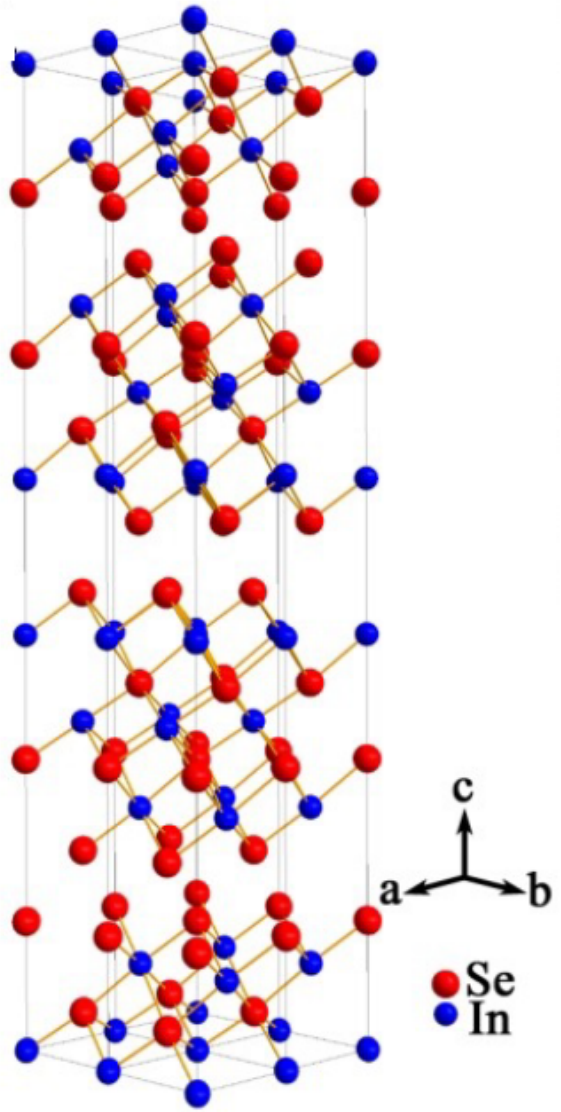



Refined experimental $In_3Se_4$

| Site | Element | Wyckoff Symbol | Symmetry | x | y | z | Occupation |
|---|---|---|---|---|---|---|---|
| In1 | In | 3a | -3m | 0.00000 | 0.00000 | 0.0000 | 1 |
| In2 | In | 6c | 3m | 0.00000 | 0.00000 | 0.4150(1) | 1 |
| Se1 | Se | 6c | 3m | 0.00000 | 0.00000 | 0.1242(1) | 1 |
| Se2 | Se | 6c | 3m | 0.00000 | 0.00000 | 0.3012(1) | 1 |

Perfect $In_3Se_4$

| Site | Element | Wyckoff Symbol | Symmetry | x | y | z | Occupation |
|---|---|---|---|---|---|---|---|
| In1 | In | 3a | -3m | 0.00000 | 0.00000 | 0.0000 | 1 |
| In2 | In | 6c | 3m | 0.00000 | 0.00000 | 0.4271 | 1 |
| Se1 | Se | 6c | 3m | 0.00000 | 0.00000 | 0.1278 | 1 |
| Se2 | Se | 6c | 3m | 0.00000 | 0.00000 | 0.2917 | 1 |



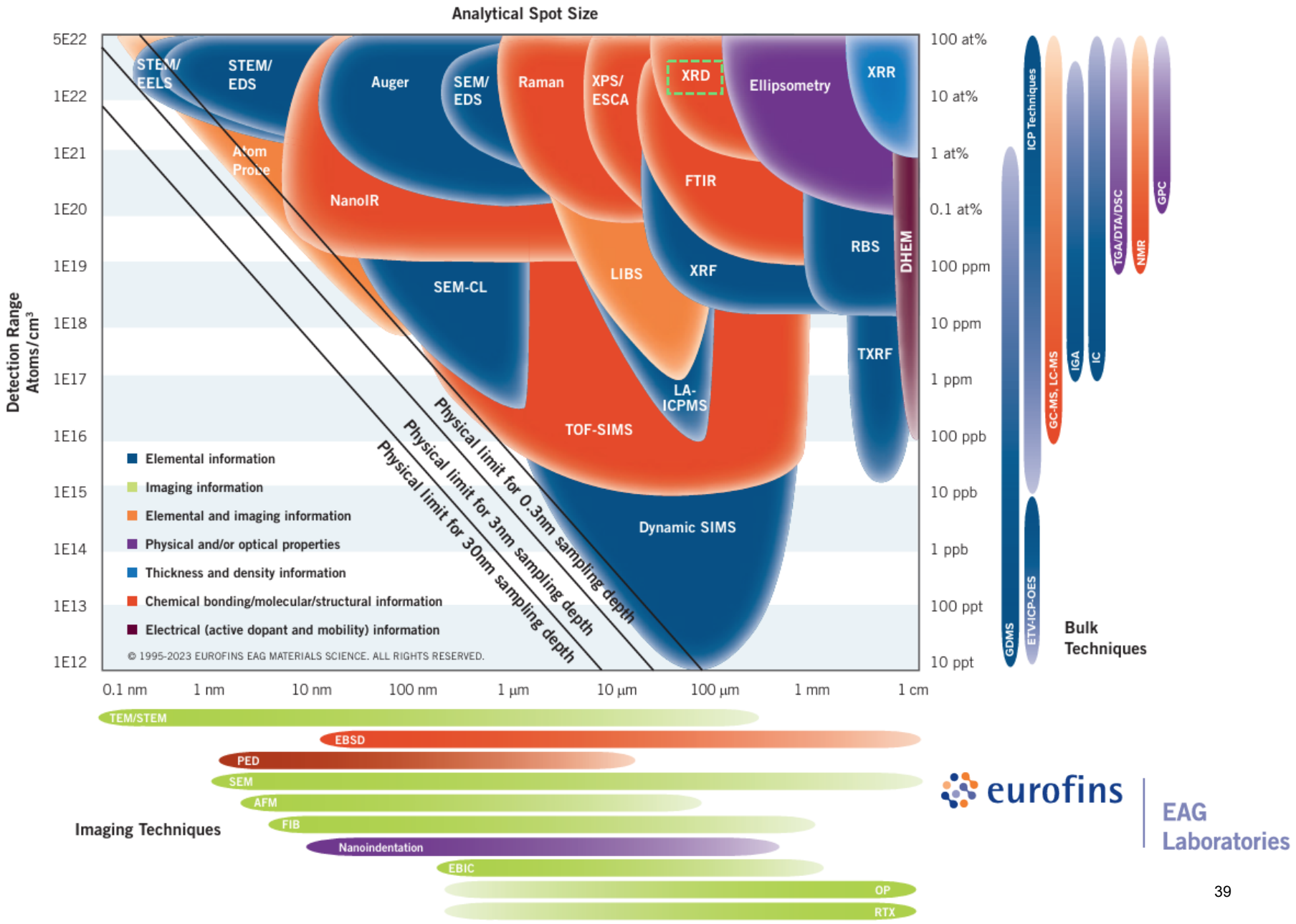
## Specifics of X-ray diffraction

- Powders, single/poly-crystalline bulks, thin films of few nm
- Sample surface must be flat, which is achievable by polishing for bulk samples.
- Surface deformation/stressing introduced by polishing should be removed by thermal annealing.
- Small crystal dimensions (*e.g.,* 100 nm or less) lead to peak broadening.
- Large crystals lead to less diffraction occurrence(10-5 nm).

## Applications of X-ray diffraction

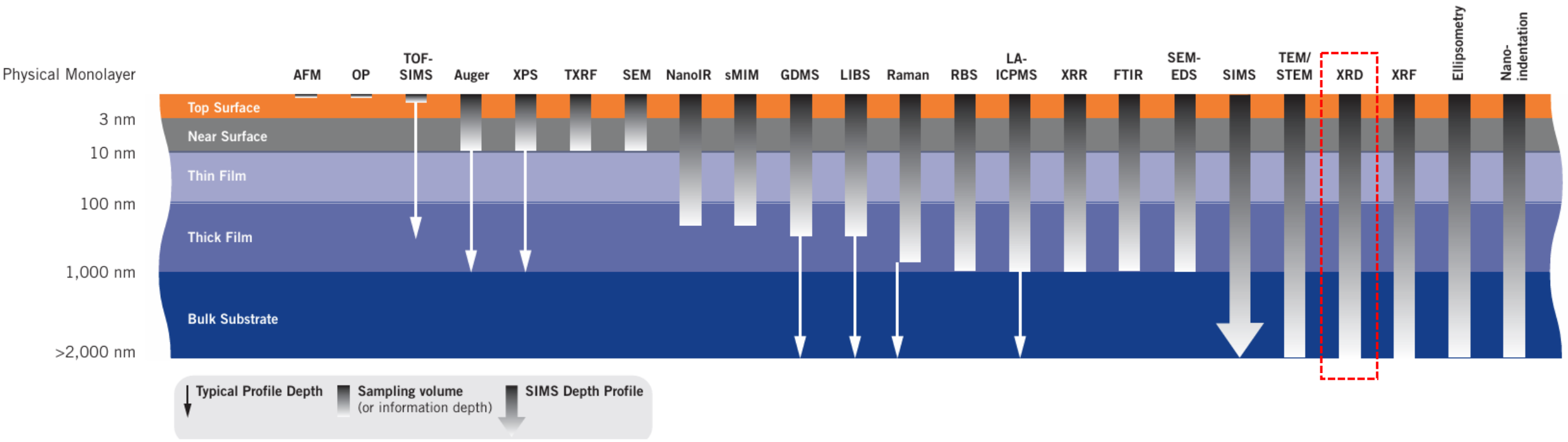
- Identification of crystalline phases and their percentage (%).
- Determination of crystal structures of new materials/structures (once it is done, it can be added to ICDD database).
- Determination of the preferred orientations of specimens (likely microstructures).
- Measurement of texture, grain size, and internal stress.
- Determination of thermal expansion and compactness of materials due to its high accuracy (10-5 nm).

## Analytical resolution *vs.* Detection limit




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# Typical analysis depth for techniques




eurofins | EAG Laboratories

**Reference & useful links:**


1. Crystallography, W. Borchardt-Ott
2. Crystallography and Crystal Defects, A. Kelly, G.W. Groves, and P. Kidd
3. The Basics of Crystallography and Diffraction, C. Hammond
4. The Structure of Materials, S.M. Allen and E.L. Thomas
5. http://www.matter.org.uk/matscicdrom/manual/xl.html
6. http://www.msm.cam.ac.uk/phasetrans/2001/intro.cryst.pdfhttps://www.bbc.co.uk/bitesize/guides/z463rwx/revision/1
7. https://www.satnanoparticles.com/news-show-2323.html
8. https://chem.libretexts.org/Bookshelves/Introductory_Chemistry/Introductory_Chemistry_%28CK-12%29/13%3A_States_of_Matter/13.14%3A_Unit_Cells
9. https://www.brainkart.com/article/Primitive-and-non-primitive-unit-cell_38660/
10. https://www.xtal.iqf.csic.es/Cristalografia/parte_04-en.html
11. SmartLab | Rigaku Global Website
12. https://myscope.training/XRD_Factors_effecting_peak_width
13. https://www.tandfonline.com/doi/full/10.1080/01411594.2017.1350958#d1e220

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# Thank you


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CRICOS code 00025B